

Maxwell Equations :-

In the beginning of the 19th century, many experimental and theoretical works had been accomplished in the understanding of electromagnets.

In 1780's Coulomb's law of electrostatic had been established. In 1825 Ampere published his Ampere's law. Michael Faraday published his law in 1834. Lenz solved the problem of the direction of induction.

James Clerk Maxwell, after his work published a series of papers from 1850s to 1870s.

In 1862, he derived the speed of light from the expression of the velocity of electromagnetic waves in relation to the vacuum constants.

The Final form of Maxwell's equations was published in 1865 in the mathematical form.

Equation 1 :-

By Gauss's law

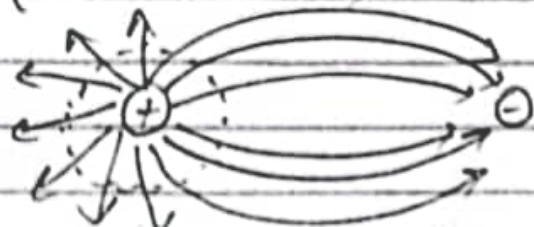
$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon_0} Q$$

This can also be written as:

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} = \frac{\rho}{\epsilon_0}$$

$$\boxed{\vec{\nabla} \cdot \vec{E} = 0} \text{ --- (1) (For vacuum)}$$

Consider the closed surface around the



charge, the divergence of $\vec{E}$ field $\neq 0$
 So this can be written as: $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon}$

Equation 2:-

By Gauss's law in terms of magnetism.

$$\phi_B = \oint \vec{B} \cdot d\vec{A} = 0$$

OR

$$\boxed{\vec{\nabla} \cdot \vec{B} = 0} \quad \text{--- (2)}$$

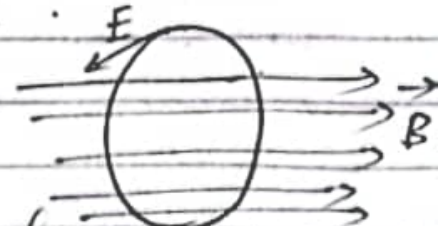

Consider the closed path around the N-pole of a bar magnet. The No. of magnetic field lines leaving is equal to the no. of magnetic field lines entering the closed surface. So net magnetic flux is zero. OR the divergence of magnetic field intensity is zero.

Equation 3:-

By Faraday's law of electromagnetic induction.

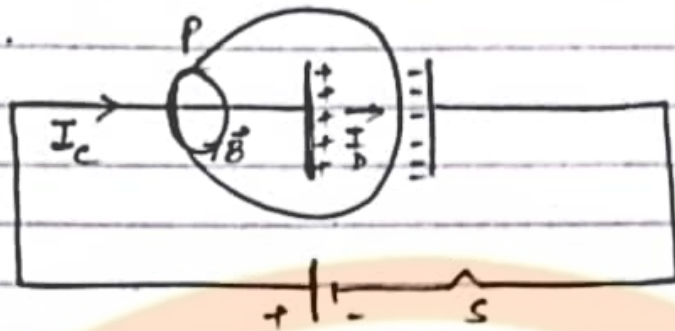
$$\mathcal{E} = \oint \vec{E} \cdot d\vec{S} = - \frac{d\phi_B}{dt} \quad \text{(-ve sign due to Lenz's law)}$$

rate of change of magnetic flux induces e. field in the closed loop. So due to change of magnetic field with time, curl of $\vec{E}$ field is produced i.e.

$$\boxed{\vec{\nabla} \times \vec{E} = - \frac{dB}{dt}} \quad \text{--- (3)}$$


Equation: 4 :-

Consider the following circuit in which a capacitor is connected to a battery and switch.



when the switch is made ON, The capacitor starts charging. During this time, current I_c flows through the loop at position P, and magnetic field is induced.

when we take the closed surface like balloon around +ve plate, due to no current I_c (conduction), field is zero. But at the same time $\vec{B} \neq 0$ and $\vec{B} = 0$ is not possible. So Maxwell consider that in the closed surface $\vec{B}$ is produced due to the displacement current I_d is from +ve to -ve plate of the capacitor. (curl of $\vec{B}$ is produced) I_d is present only when I_c is present.

By Ampere's circuital Law

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \vec{J}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

$$\Rightarrow \vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}) = \mu_0 \vec{\nabla} \cdot \vec{J}$$

$$\Rightarrow 0 = \mu_0 \vec{\nabla} \cdot \vec{J}$$

$$\Rightarrow \vec{\nabla} \cdot \vec{J} = 0$$

$$\text{But } \vec{\nabla} \cdot \vec{J} \neq 0$$

$$\text{So } \oint \vec{B} \cdot d\vec{s} = \mu_0 (I_c + I_d)$$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_c + \mu_0 I_d$$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_c + \mu_0 \epsilon \frac{d\phi_E}{dt}$$

$$\boxed{\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}_c + \mu_0 \epsilon \frac{\partial \vec{E}}{\partial t}} \quad (4)$$

Electromagnetic Waves:-

An electromagnetic wave is a varying electric and magnetic field.

The em waves are travelling (not standing) transverse waves. The displacement (field strength) is perpendicular to the direction of travel. em waves do not require any material medium for their propagation as electric and magnetic field both exist in space. The em waves can be produced in the two ways:

- i) A moving charge is a current, which produces a magnetic field. An accelerating charge means a current that is changing its value, which induces a changing magnetic field in addition to a changing electric field.
- ii) When an electron from excited state moves to a lower energy state, it releases energy as an e.m wave.

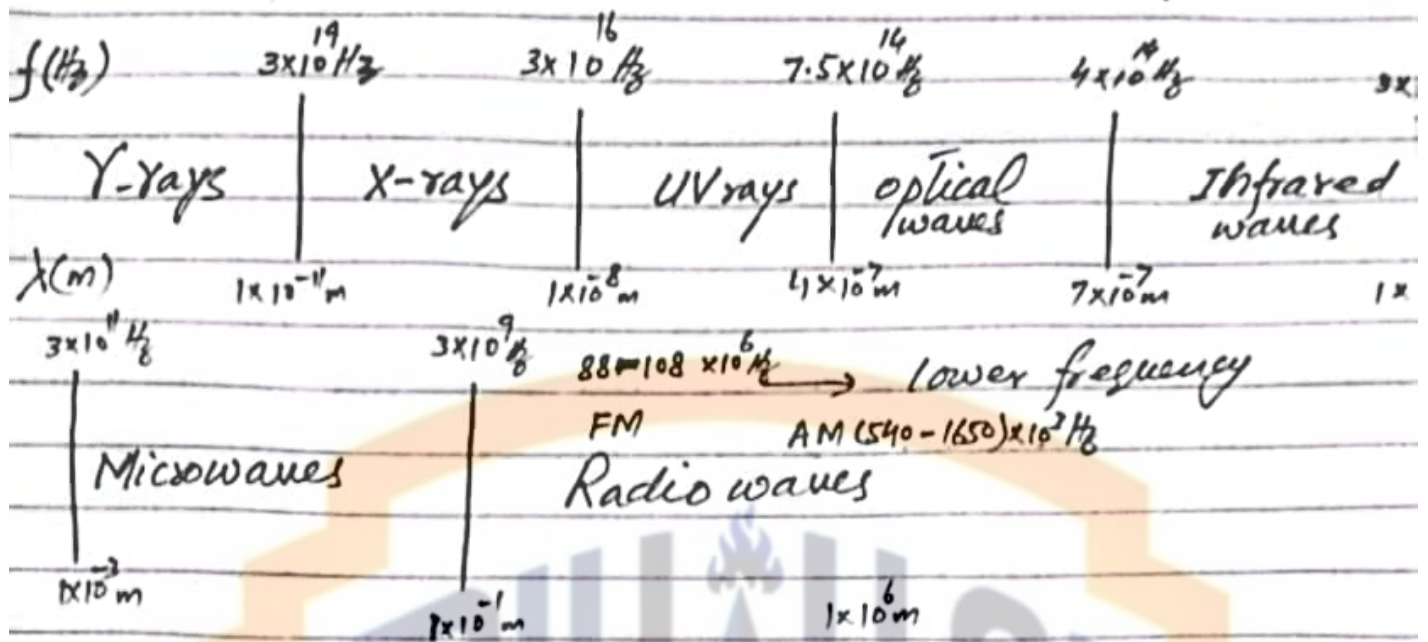
The changing electric field induces a magnetic field and changing magnetic field induces an electric field.

So when you create a change in an electric or magnetic field, you create a self-sustaining wave that travels away in all directions.

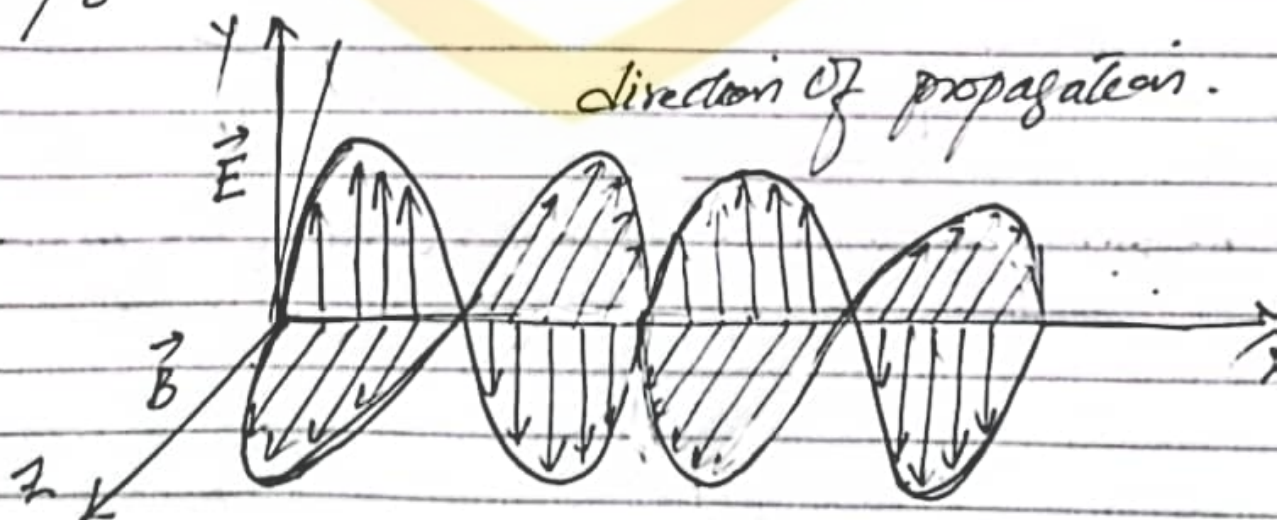
There are different types of em waves.

The only physical difference between different types of EM waves is their wave length & frequency & as a result their energy. We can organize the types on the bases of

these properties on a chart called em spectrum.



James Clerk Maxwell in 1873, unified electricity & magnetism with his famous equations and showed that light is an em wave. But this theory was not accepted at that time. Fifteen years later, Heinrich Hertz experimentally justified the Maxwell's theory. He showed that em waves could be generated by oscillating charges (en). He showed that em waves are transverse in which $\vec{E}$ & $\vec{B}$ are $\perp$ to each other & to the direction of propagation.



The wave equation is given as:

$$\nabla^2 U(x,t) = \frac{1}{c^2} \frac{\partial^2 U(x,t)}{\partial t^2}$$

OR

$$\nabla^2 U - \frac{1}{c^2} \frac{\partial^2 U}{\partial t^2} = 0 \quad \text{--- (1)}$$

∇^2 is Laplacian operator $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

The Maxwell's equations are given as:

$$\nabla \cdot E = \rho/\epsilon_0 \Rightarrow \nabla \cdot E = 0 \quad \text{(in space)}$$

$$\nabla \cdot B = 0 \quad \text{--- (ii)}$$

$$\nabla \times E = -\frac{\partial B}{\partial t} \quad \text{--- (iii)}$$

$$\nabla \times B = \mu_0 \epsilon_0 \frac{\partial E}{\partial t} + \mu_0 J \quad \text{--- (iv)}$$

Since Taking curl of eq. (iii)

$$\nabla \times (\nabla \times E) = -\nabla \times \frac{\partial B}{\partial t}$$

$$\nabla \times (\nabla \times E) = -\frac{\partial}{\partial t} (\nabla \times B)$$

$$\nabla \times (\nabla \times E) = -\frac{\partial}{\partial t} \left(\mu_0 \epsilon_0 \frac{\partial E}{\partial t} \right) \quad \text{Using eq. (iv)}$$

$$\nabla (\nabla \cdot E) - (\nabla \cdot \nabla) E = -\mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}$$

$$\nabla (0) - \nabla^2 E = -\mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}$$

$$0 - \nabla^2 E = -\mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}$$

$$-\nabla^2 E = -\mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}$$

$$\nabla^2 E = -\mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}$$

$$\nabla^2 E - \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2} = 0 \quad \text{--- (2)}$$

$$\left\{ \begin{array}{l} \text{As} \\ \nabla \times (\nabla \times E) \\ = B(A \cdot C) - C(A \cdot B) \\ \text{So} \\ \nabla \times (\nabla \times E) \\ = \nabla (\nabla \cdot E) - E (\nabla \cdot \nabla) \end{array} \right.$$

By taking the curl of eq. (iv), we have the same equation as eq. (2) for magnetic field vector.

$$\nabla^2 B - \mu_0 \epsilon_0 \frac{\partial^2 B}{\partial t^2} = 0 \quad \text{--- (3)}$$

On comparing eq (1) with eq (2) & eq (3)

$$\frac{1}{c^2} = \mu_0 \epsilon_0$$

$$\Rightarrow c^2 = \frac{1}{\mu_0 \epsilon_0}$$

$$\Rightarrow c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

For free space (vacuum);

$$\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2 \quad \& \quad \epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

putting these values in above relation.

$$c = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.85 \times 10^{-12}}}$$

$$c = 2.998 \times 10^8 \text{ m/s}$$

$$c = 3 \times 10^8 \text{ m/s}$$

For medium other than space; Speed of em waves is given as:

$$v = \frac{1}{\sqrt{\mu \epsilon}}$$

$$v = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \cdot \frac{1}{\sqrt{\mu_r \epsilon_r}}$$

$$v = c \cdot \frac{1}{\sqrt{\mu_r \epsilon_r}} = \frac{c}{\sqrt{\mu_r \epsilon_r}}$$

$$v = \frac{c}{\sqrt{\epsilon_r}} \quad (\text{For Non-ferromagnetic substance; } \mu_r = 1)$$

$$\text{as } \sqrt{\epsilon_r} > 1 \text{ so } v < c$$

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Also by definition of refractive Index:

$$n = \frac{c}{v}$$

$$v = \frac{c}{n}$$

as $n > 1$ for medium other than vacuum,

So

$$v < c$$

Hence speed of light decreases in material medium.

Energy density of Electric field

As For Capacitor.

$$V = \frac{W}{Q} = \frac{Fd}{Q} = \frac{QEd}{Q} = Ed$$

$$\boxed{V = Ed} \quad \text{--- (i)}$$

Also by Gauss's law

$$\phi_E = E \cdot A = \frac{Q}{\epsilon_0}$$

$$\Rightarrow E = \frac{Q}{\epsilon_0 A}$$

$$\Rightarrow \boxed{Q = \epsilon_0 EA} \quad \text{--- (ii)}$$



initially $V_0 = 0$
finally $V_{\text{max}} = V$

$$V_{\text{av}} = \frac{0 + V}{2} = \frac{V}{2}$$

Now

$$\text{As } W = Q V_{\text{av}}$$

$$\Rightarrow W = (\epsilon_0 EA) \left(\frac{V}{2} \right) \text{ using (i)}$$

$$= \frac{1}{2} (\epsilon_0 EA) (V)$$

$$= \frac{1}{2} (\epsilon_0 EA) (Ed) \text{ using (ii)}$$

$$= \frac{1}{2} \epsilon_0 (Ad) E^2$$

$$W = \frac{1}{2} \epsilon_0 E^2 (\text{Volume b/w plates})$$

$$\frac{W}{\text{vol}} = \frac{1}{2} \epsilon_0 E^2$$

$$\text{or } \frac{U}{V} = \frac{1}{2} \epsilon_0 E^2$$
$$\boxed{U_F = \frac{1}{2} \epsilon_0 E^2}$$

Energy density of magnetic field.

In case of Inductor

$$\mathcal{E} = L \frac{\Delta I}{\Delta t}$$

As

$$W = QV$$

$$W = Q\mathcal{E}$$

$$W = Q L \frac{\Delta I}{\Delta t}$$

$$W = L \frac{\Delta Q}{\Delta t} I$$

$$W = L \left(\frac{1}{2} I \right) I$$

$$W = \frac{1}{2} L I^2 \quad (1)$$

As $\mathcal{E} = N \frac{\Delta \phi}{\Delta t}$ — (i)

$\mathcal{E} = L \frac{\Delta I}{\Delta t}$ — (ii)

$\Delta I = I$

from (i) & (ii)

$$N \frac{\Delta \phi}{\Delta t} = L \frac{\Delta I}{\Delta t}$$

$$\frac{\Delta}{\Delta t} (N \phi) = \frac{\Delta}{\Delta t} (LI)$$

$$N \phi = LI$$

↓

$$N B \cdot A = LI$$

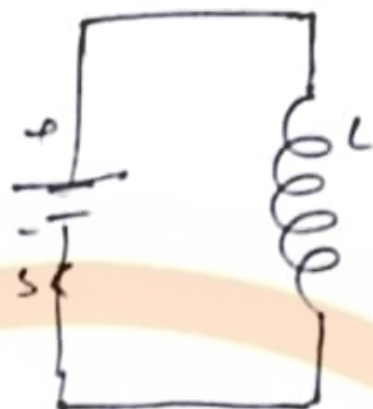
$$N \cdot \mu_r \mu_0 n I \cdot A = LI$$

$$n L \mu_r \mu_0 n I / A = LI$$

$$n^2 L \mu_r \mu_0 A = L$$

putting these values in eq (1)

$$W = \frac{1}{2} (n^2 L \mu_r \mu_0 A) \left(\frac{B}{\mu_r \mu_0 n} \right)$$



$$I_{\min} = 0$$

$$I_{\max} = I$$

$$I_{\text{av}} = \frac{\Delta I}{\Delta t} = \frac{I + 0}{2}$$

$$\frac{\Delta \phi}{\Delta t} = I/2$$

$$\Delta I = I_{\max} - I_{\min}$$

$$\Delta I = I - 0$$

As $n = \frac{N}{l}$

$$N = n l$$

$$W = \frac{1}{2} n^2 \mu_r \mu_0 l A \frac{B^2}{\mu_r \mu_0 n^2}$$

$$W = \frac{1}{2} \frac{B^2}{\mu_0} (l A)$$

$$W = \frac{1}{2} \frac{B^2}{\mu_0} (\text{Vol})$$

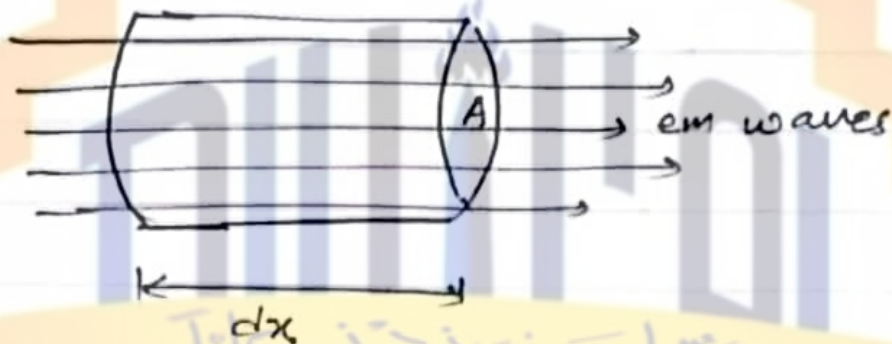
$$\frac{W}{\text{Vol}} = \frac{1}{2} \frac{B^2}{\mu_0}$$

$$U_B = \frac{1}{2} \frac{B^2}{\mu_0}$$

Intensity of EM Waves:-

It is defined as the energy of em waves crossing per second per unit area normally to wave propagation is called the intensity of em waves. It is denoted by I . Its unit is $J/s\,m^2$ or W/m^2 .

$$I = \frac{E}{A \Delta t} \quad \text{--- (1)}$$



Consider a cylindrical volume in the path of em waves as shown. The area of cross-section is A and length element is $dx = c \Delta t$. So volume becomes $V = A \times dx = A c \Delta t$.

By definition the energy density is given as:

$$u_E = \frac{U}{V}$$

OR $U = u_E V$

$$U = u_E A c \Delta t$$

putting this value in eq (1), we have; (where $U = E$ energy)

$$I = \frac{u_E A c \Delta t}{A \Delta t}$$

$$I = u_E c$$

As we know that the energy of E wave field vector $\vec{E}$ is given as

$$(U_E)_E = \frac{1}{2} \epsilon_0 E^2$$

& that of magnetic field vector $\vec{B}$ is given as

$$(U_B)_B = \frac{1}{2} \frac{B_0^2}{\mu_0}$$

Now for $\vec{E}$

$$I = \frac{1}{2} \epsilon_0 E_0^2 C$$

& for $\vec{B}$

$$I = \frac{1}{2} \frac{B_0^2}{\mu_0} C$$

Total intensity of EM waves will be

$$I = \left(\frac{1}{2} \epsilon_0 E_0^2 + \frac{1}{2} \frac{B_0^2}{\mu_0} \right) C$$

Poynting Vector:-

The rate of energy flow through a unit area normally to the propagation of e.m waves is called poynting vector. It is denoted by $\vec{S}$. Its unit is W/m^2

If dU energy passes through unit area dA in time interval dt , then

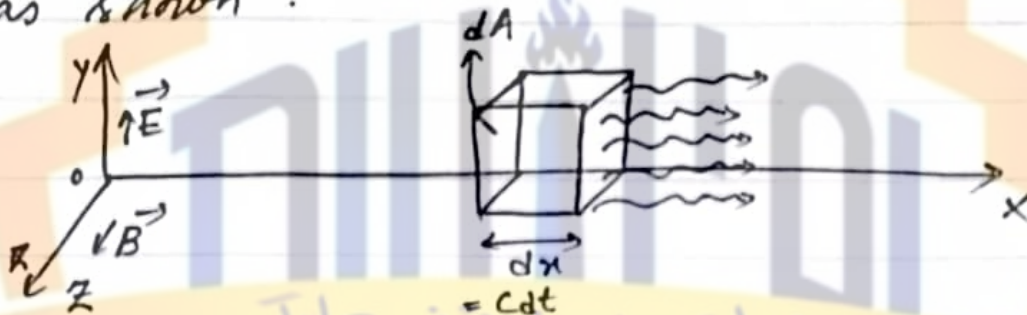
$$\vec{S} = \frac{dU}{dA dt}$$

In vector form

$$\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0}$$

To prove this relation; Consider e.m waves propagating along x-axis.

Take a cubic volume around the waves with facing area dA . The waves cross this volume element by covering distance dx as shown.



A3 Energy density = $\frac{\text{energy}}{\text{unit volume}}$

$$U = \frac{dU}{dV}$$

$$dU = U dV \quad \text{--- (i)}$$

where $U = U_E + U_B = \frac{1}{2} \epsilon_0 E^2 + \frac{B^2}{2\mu_0} \quad \text{--- (ii)}$

A3 $c = \frac{E}{B} \Rightarrow \boxed{E = cB} \quad \& \quad \boxed{B = \frac{E}{c}}$

putting these values eq (ii) becomes as:

$$U = \frac{1}{2} \epsilon_0 c B E + \frac{1}{2\mu_0 c} B E$$

$$U = \frac{1}{2\mu_0 c} B E + \frac{1}{2\mu_0 c} B E$$

$$\boxed{U = \frac{BE}{\mu_0 c}}$$

$$\left\{ \begin{aligned} c^2 &= \frac{1}{\mu_0 \epsilon_0} \\ \epsilon_0 c &= \frac{1}{\mu_0} \end{aligned} \right.$$

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putting this value in eq ①

$$\Rightarrow dU = \frac{EB}{\mu_0 c} dV$$

$$\left\{ \begin{aligned} dV &= dA dx \\ &= dA c dt \end{aligned} \right.$$

$$dU = \frac{EB}{\mu_0 c} dA c dt$$

$$dU = \frac{EB}{\mu_0} dA dt$$

$$\frac{dU}{dA dt} = \frac{EB}{\mu_0}$$

$$\vec{S} = \frac{EB \sin 90^\circ}{\mu_0}$$

$$(As \sin 90^\circ = 1)$$

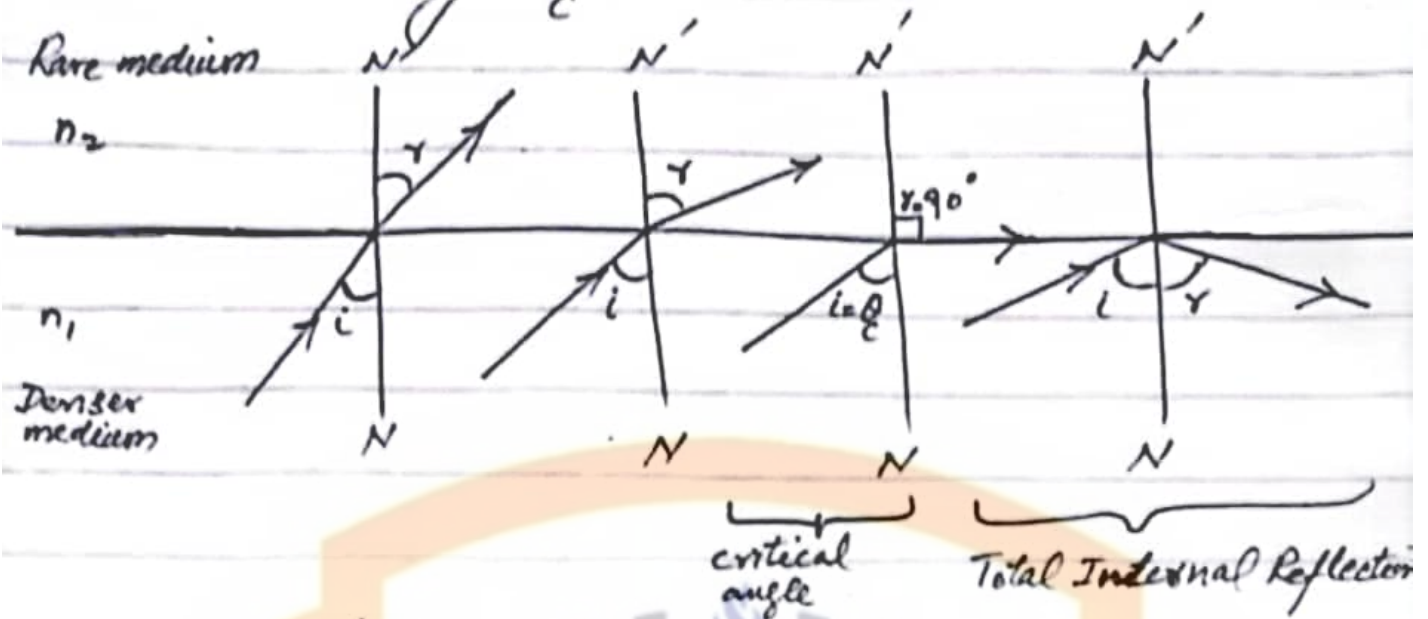
$$\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0} \quad \text{or} \quad \vec{S} = \vec{E} \times \vec{H}$$

Total Internal Reflection

when the em wave enters from denser into rare medium, then it bends away from the normal. By increasing the incident angle i the refracted angle r also increases.

On further increasing the incident i , the corresponding refracted angle becomes perpendicular with normal. Such an incident angle in the denser medium whose refracted angle becomes 90° , is called critical angle

denoted by " θ_c ".



Consider a denser medium with refractive index n_1 & a rare medium n_2 & ray of em waves enter from denser into rare medium as shown. Then by Snell's law

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n_1 \sin i = n_2 \sin r$$

$$\boxed{\frac{\sin i}{\sin r} = \frac{n_2}{n_1}}$$

Similarly for critical angle $\theta_c = i$, $r = 90^\circ$. Then Snell's law becomes as:

$$\frac{\sin \theta_c}{\sin 90^\circ} = \frac{n_2}{n_1}$$

$$\Rightarrow \boxed{\sin \theta_c = \frac{n_2}{n_1}}$$

Acceptance angle:-

In optical fibre, The maximum angle of incidence at the interface of air medium and core medium for which the light ray enters into the core & travels along the interface of core & cladding, is called the acceptance angle, denoted by " θ_0 ".

Let n_0 = refractive index of air

n_1 = refractive index of core

n_2 = refractive index of cladding

Then by Snell's law at point A,

$$n_0 \sin \theta_0 = n_1 \sin \theta_1$$

$$\sin \theta_0 = \frac{n_1}{n_0} \sin \theta_1 \quad \text{--- (1)}$$

By Snell's law at point B

$$n_1 \sin (90 - \theta_1) = n_2 \sin 90^\circ$$

$$n_1 \cos \theta_1 = n_2$$

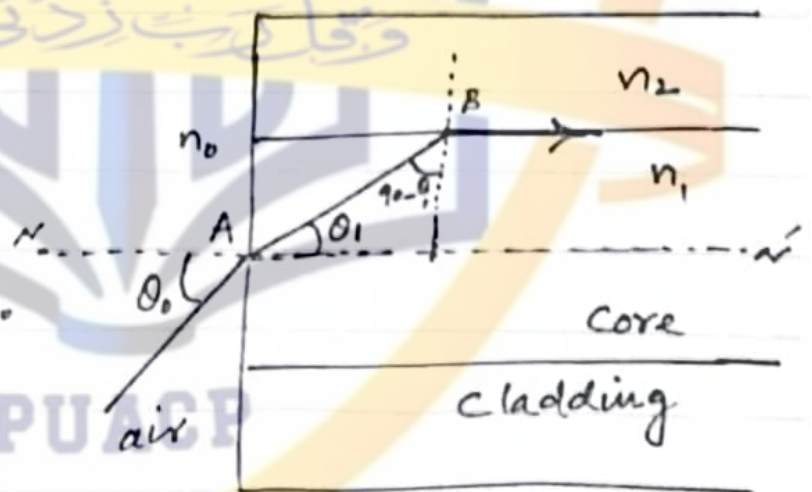
$$\cos \theta_1 = \frac{n_2}{n_1} \quad \text{--- (2)}$$

$$\text{As } \sin \theta_1 = \sqrt{1 - \cos^2 \theta_1}$$

$$\sin \theta_1 = \sqrt{1 - \frac{n_2^2}{n_1^2}}$$

putting this value in eq (1)

$$\Rightarrow \sin \theta_0 = \frac{n_1}{n_0} \sqrt{1 - \frac{n_2^2}{n_1^2}}$$



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$$\sin \theta_c = \frac{n_1}{n_0} \frac{\sqrt{n_1^2 - n_2^2}}{n_1}$$

$$\sin \theta_c = \frac{1}{n_0} \sqrt{n_1^2 - n_2^2}$$

As for air, $n_0 = 1$

$$\sin \theta_c = \sqrt{n_1^2 - n_2^2}$$

$$\theta_c = \sin^{-1} \left[\sqrt{n_1^2 - n_2^2} \right]$$

So by knowing the refractive index of core & cladding, the acceptance angle θ_c can be calculated.

Total internal reflection technique is used in periscope, endoscope, optical fibre, Binocular etc.

Reflection & Refraction of EM Waves.

Reflection of e.m waves/light:-

When the ray of light (e.m waves) is incident on the boundary of a medium and bounces back, the phenomenon is called reflection of light. (e.g. in case of mirror)

Refraction of e.m waves/light:-

When the ray of light enters from one medium into another medium, it deviates from

its path, such bending of light in the second medium is called refraction of light.

Following are the laws of Reflection & Refraction of light:

i) Incident ray, reflected ray & refracted ray all lie in the same plane.

ii) The frequency of e.m waves which are incident, reflected & refracted are same.

iii) incident angle = reflected angle
OR angle of incidence = angle of reflection
i.e. $\angle i = \angle r$

iv) Snell's law:

$$n_1 \sin \theta_i = n_2 \sin \theta_r$$

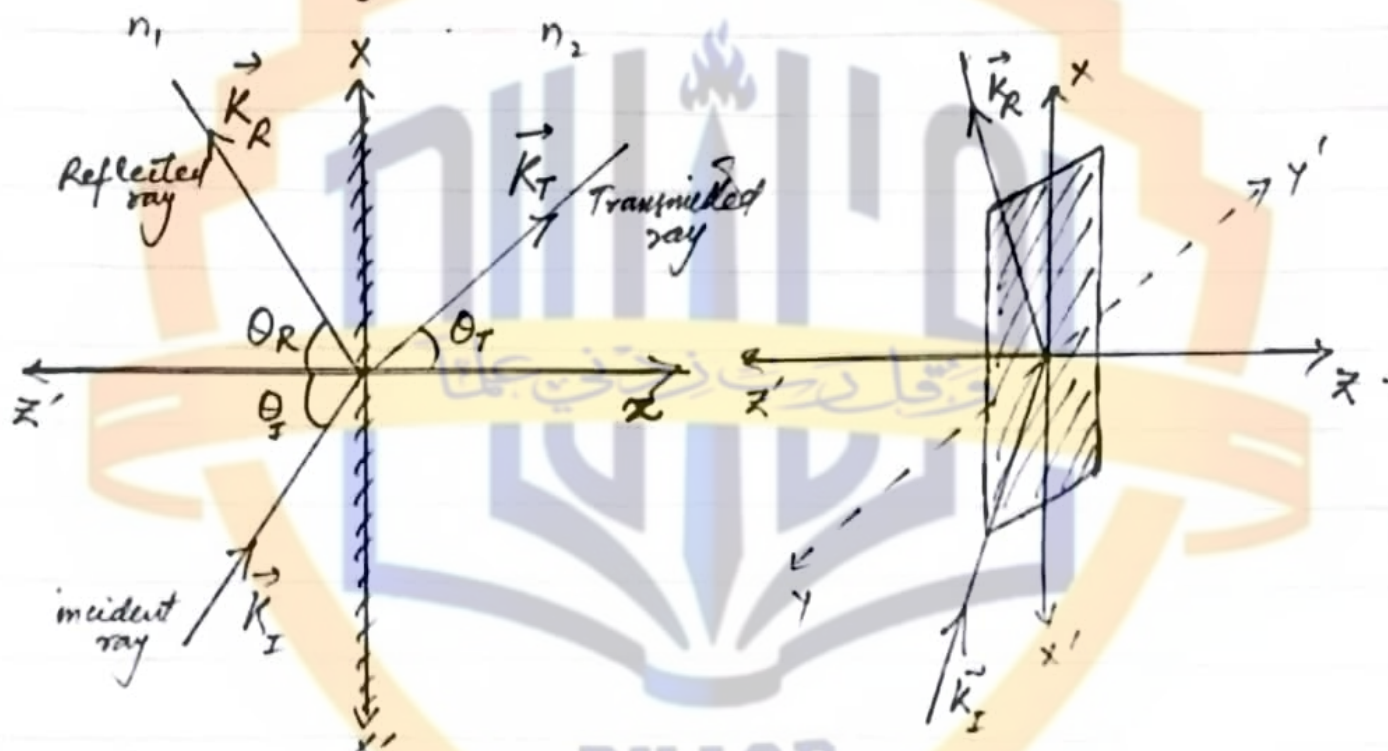
where n_1, n_2 are refractive index of medium 1 & 2.
 θ_i = incident angle
 θ_r = refracted angle.

Now To prove these laws of reflection & refraction related to E.M waves by considering the following ray diagram (we consider the E-field of em waves)
let K_i be the Electric wave vector of incident ray
 K_r be the electric wave vector of reflected ray

K_T be the electric wave vector of Transmitted (refracted ray).

$\omega_I, \omega_R, \omega_T$ be the ^{angular} frequencies of incident, reflected & refracted rays.

$\theta_I, \theta_R, \theta_T$ be the angles made by incident, reflected & refracted ray with (z-axis) normal of the two mediums.



As by wave equation

$$\nabla^2 E - \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2} = 0$$

The solution of this equation is given as:

$$\vec{E} = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

The Solution

for incident ray	$\vec{E}_I = E_{0I} e^{i(\vec{k}_I \cdot \vec{r} - \omega_I t)}$	— (1)
for Reflected ray	$\vec{E}_R = E_{0R} e^{i(\vec{k}_R \cdot \vec{r} - \omega_R t)}$	— (2)
for Refracted ray	$\vec{E}_T = E_{0T} e^{i(\vec{k}_T \cdot \vec{r} - \omega_T t)}$	— (3)

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(i) Applying the boundary condition across the interface of two mediums.

$$\sum E_{1t} = \sum E_{2t} \quad \text{or } \sum (E_1)_{||} = \sum (E_2)_{||}$$

$$\Rightarrow \left(\vec{E}_I \right)_{||} e^{i(\vec{k}_I \cdot \vec{r} - \omega_I t)} + \left(\vec{E}_R \right)_{||} e^{i(\vec{k}_R \cdot \vec{r} - \omega_R t)} = \left(\vec{E}_T \right)_{||} e^{i(\vec{k}_T \cdot \vec{r} - \omega_T t)} \quad \text{--- (1)}$$

As the boundary condition, the tangential components do not depend on the position of interface & time, so exponential on R.H.S common.

$$\Rightarrow \left(\vec{k}_I \cdot \vec{r} \right)_{z=0} = \left(\vec{k}_R \cdot \vec{r} \right)_{z=0} = \left(\vec{k}_T \cdot \vec{r} \right)_{z=0} \quad \begin{cases} \vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \\ \vec{k}_I = k_{Ix}\hat{i} + k_{Iy}\hat{j} + k_{Iz}\hat{k} \\ z=0 \end{cases}$$

$$(k_{Ix})x + (k_{Iy})y = (k_{Rx})x + (k_{Ry})y = (k_{Tx})x + (k_{Ty})y$$

It is possible only if:

$$\Rightarrow \begin{aligned} k_{Ix} &= k_{Rx} = k_{Tx} & \text{--- (i)} & \left\{ \begin{array}{l} \text{As incident ray} \\ \text{is in } x-z \text{ plane} \\ \text{ie } y \text{ component} = 0 \\ \text{ie } k_{Iy} = 0 \end{array} \right. \\ k_{Iy} &= k_{Ry} = k_{Ty} & \text{--- (ii)} & \end{aligned}$$

$$\text{So } k_{Iy} = k_{Ry} = k_{Ty} = 0 \quad (x-z \text{ plane})$$

It means k_I, k_R, k_T all lies in same $(x-z \text{ plane})$

ii) From equation (1) $\omega_I t = \omega_R t = \omega_T t$

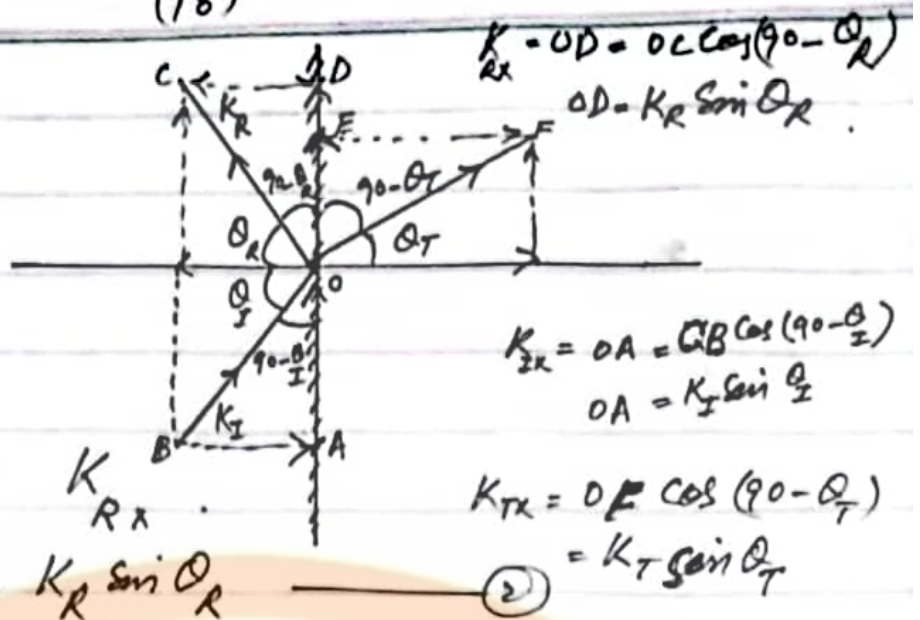
$$\Rightarrow \omega_I = \omega_R = \omega_T$$

$$\text{OR } 2\pi\nu_I = 2\pi\nu_R = 2\pi\nu_T \Rightarrow \nu_I = \nu_R = \nu_T$$

It means frequency of incident, reflected & refracted rays remains equal.

(18)

(iii)



As $K_{Ix} = K_{Rx}$
 $K_I \sin \theta_I = K_R \sin \theta_R$ — (2)

As $K_I = \frac{w_I}{v_I}$, $K_R = \frac{w_R}{v_I}$

eq (2) becomes as;

$$\frac{w_I \sin \theta_I}{v_I} = \frac{w_R \sin \theta_R}{v_I} \quad (w_I = w_R \text{ proved})$$

$$\Rightarrow \sin \theta_I = \sin \theta_R$$

$$\Rightarrow \theta_I = \theta_R$$

$\Rightarrow$ Angle of incidence = Angle of Reflection
 (proved)

(iv)

As by eq (i)

$$K_{Ix} = K_{Tx}$$

$$K_I \sin \theta_I = K_T \sin \theta_T$$

$$\frac{w_I \sin \theta_I}{v_I} = \frac{w_T \sin \theta_T}{v_2} \quad (w_I = w_T \text{ proved})$$

$$\frac{1}{v_I} \sin \theta_I = \frac{1}{v_2} \sin \theta_T \quad \text{--- (3)}$$

As we know that refractive index of

a medium is given as:

$$n = \frac{c}{v} \Rightarrow v = \frac{c}{n}$$

$$\Rightarrow v_1 = \frac{c}{n_1} \quad \& \quad v_2 = \frac{c}{n_2}$$

putting these values in eq (3)

$$\Rightarrow \frac{1}{c/n_1} \sin \theta_i = \frac{1}{c/n_2} \sin \theta_r$$

$$\frac{n_1}{c} \sin \theta_i = \frac{n_2}{c} \sin \theta_r$$

$$n_1 \sin \theta_i = n_2 \sin \theta_r$$

OR $n_1 \sin \angle i = n_2 \sin \angle r$

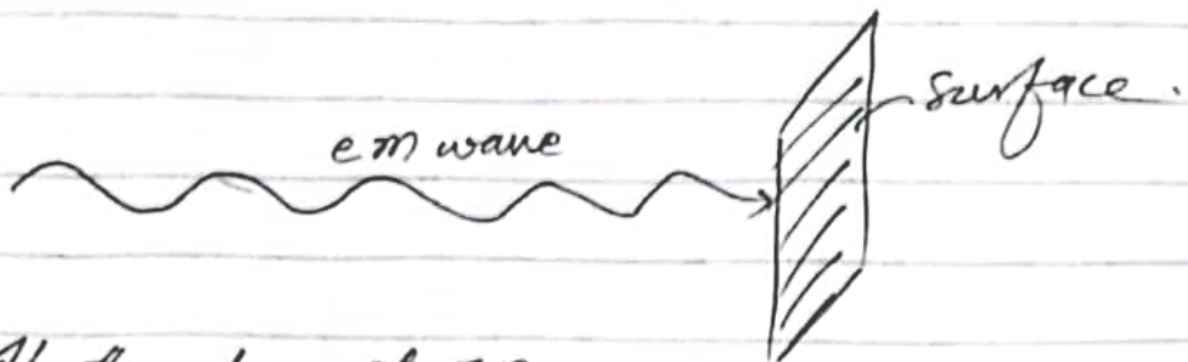
which is the required form of Snell's law proved.

Radiation Pressure of EM Waves:-

Since the EM waves carry energy, so they also carry momentum as well.

When an EM wave hits the surface of an object, it will exert force on the surface like a solid object.

So the radiation pressure can be defined as the force exerted by EM wave per unit area of some object. It is denoted by p & having unit N/m^2 .



If the beam of EM waves is completely absorbed by the object, then momentum absorbed is given as $\Delta p = \frac{\Delta U}{c}$

If the beam of EM waves is completely reflected by the surface of the object then change in momentum of the wave is $\Delta p = \frac{2\Delta U}{c}$

where ΔU is the energy of the wave absorbed in time Δt .

Now

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

$$P = \frac{F}{A}$$

$$P = \frac{\Delta p}{\Delta t} \cdot \frac{1}{A}$$

$$P = \frac{\Delta U/c}{\Delta t A} = \left(\frac{\Delta U}{\Delta t A} \right) / c \quad \text{--- (1)}$$

As

$$W = Fd$$

$$\Delta U = F \Delta x$$

$$\Rightarrow F = \Delta U / \Delta x$$

Also we know that

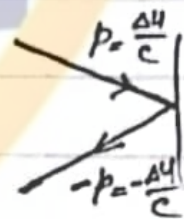
$$F = \frac{\Delta p}{\Delta t}$$

$$\Delta p = F \Delta t$$

$$\Rightarrow \Delta p = \frac{\Delta U}{\Delta x} \cdot \Delta t$$

$$\Delta p = \frac{\Delta U}{\Delta x / \Delta t}$$

$$\Delta p = \frac{\Delta U}{c}$$



$$\Delta p = p_f - p_i$$

$$\Delta p = \frac{-\Delta U}{c} - \frac{\Delta U}{c} = -\frac{2\Delta U}{c}$$

$$\text{mag: of } \Delta p = 2\Delta U/c$$

(21)

We know that the relation for Poynting vector is given as,

$$\vec{S} = \frac{dU}{dt dA} \quad \text{or} \quad S = \frac{\Delta U}{\Delta t A}$$

So eq (1) can be written as:

$$\boxed{P = S/c}$$

It is the relation for radiation pressure, when EM wave is totally absorbed.

Also

$$P = \frac{F}{A} = \frac{\Delta p \cdot 1}{\Delta t A} = \frac{2 \Delta U / c}{\Delta t A} = 2 \left(\frac{\Delta U}{\Delta t A} \right) / c$$

$$\boxed{P = \frac{2S}{c}}$$

It is the relation for radiation pressure for EM wave when the wave is totally reflected.

Polarization of EM Waves:-

Unpolarized light:-

The type of e.m waves or light which has vibrations of electric & magnetic field in all directions is called unpolarized light.

eg light from sun, flame etc.



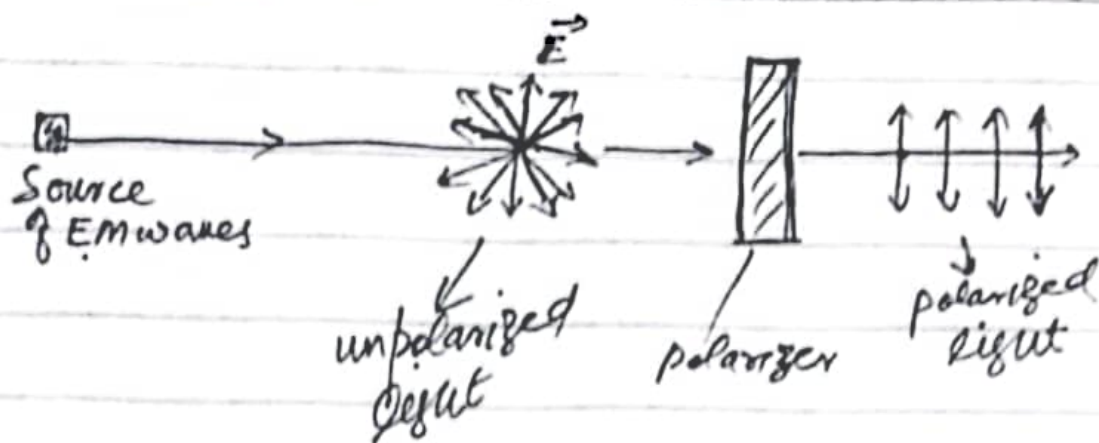
Polarized light:-

The light whose vibration of E-field vector is along the single axis (single plane) & at a fixed angle with the direction of propagation of wave is called polarized light.

Unpolarized light can be converted to the polarized light by passing through polarizing filters or discs like nicol prism.

Polarization:-

The process of restricting the vibrations of $\vec{E}$ & $\vec{B}$ vector of EM waves from many direction to a single direction (single plane) is called polarization of light / e.m waves



There are different types of polarization of EM waves; which are.

- i) plane Polarization / linear polarization.
- ii) Circular Polarization.
- iii) Elliptical Polarization.

$\vec{E}$ & $\vec{B}$ vectors are perpendicular to each other & to the direction of propagation of wave.

Here we consider only $\vec{E}$ because $\vec{E}$ field is many times stronger than $\vec{B}$ ($E = 377B$)

As $c = \frac{E}{B} \Rightarrow E = cB$.

Let the wave is propagating along z-axis.

$\vec{E}$ remains along the straight line as a function of time.

1) Linear Polarization:-

It has three cases.

- a) when x-component of E-field is present & y-component of E-field is absent, & wave propagates along z-axis. i.e.

$$E_x = \text{present} \quad E_y = 0$$

Such polarization is called Horizontal polarization of em waves.

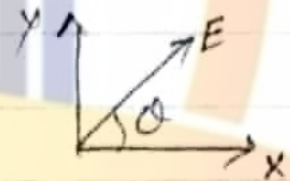
- b) If y -component of E -field is present & x -component is absent & wave propagates along z -axis, i.e. $E_x = 0, E_y = \text{present}$

Such polarization is called vertical polarization of em waves.

- c) If E_x and E_y are present and the E -field oscillates at some angle with x -axis then such polarization is called θ -polarization.

where $\theta = \tan^{-1} \left(\frac{E_y}{E_x} \right)$

$$E = \sqrt{E_x^2 + E_y^2}$$



Consider two E -field vectors $\vec{E}_1$ & $\vec{E}_2$ oscillating perpendicular to each other. The Electric field vector as a function of position & time is given as:

$$\vec{E}(z, t) = \vec{E}_0 e^{i(\omega t - kz)} \quad \text{--- (1)}$$

where $\vec{E}_0 = E_1 \hat{i} + iE_2 \hat{j}$

E_1 & E_2 can also be the magnitudes of E in x -axis & y -axis respectively.

Eq (1) can be written as:

$$\vec{E}(z, t) = (E_1 \hat{i} + i E_2 \hat{j}) e^{i(\omega t - kz)}$$

Let $z=0$

$$\vec{E}(z, t) = (E_1 \hat{i} + i E_2 \hat{j}) [\cos(\omega t - kz) + i \sin(\omega t - kz)]$$

$$\vec{E}(0, t) = (E_1 \hat{i} + i E_2 \hat{j}) (\cos \omega t + i \sin \omega t)$$

$$\vec{E}(0, t) = E_1 \cos \omega t \hat{i} + i \sin \omega t E_1 \hat{j} + i E_2 \cos \omega t \hat{j} + i^2 E_2 \sin \omega t \hat{j}$$

$$\vec{E}(0, t) = [E_1 \cos \omega t \hat{i} - E_2 \sin \omega t \hat{j}] + i [E_1 \sin \omega t \hat{i} + E_2 \cos \omega t \hat{j}]$$

Consider the real part of above equation

$$\tilde{E}(0, t) = E_1 \cos \omega t \hat{i} - E_2 \sin \omega t \hat{j}$$

$$\tilde{E}_x = E_1 \cos \omega t, \quad \tilde{E}_y = -E_2 \sin \omega t$$

$$\vec{E}(0, t) = \tilde{E}_x \hat{i} + \tilde{E}_y \hat{j}$$

$\tilde{E}_x$ & $\tilde{E}_y$ are at a phase difference of 90°

There are two cases. ⁽²⁾ * Circular Polarization

i) when $E_1 = E_2 = E$ (say)

$$\Rightarrow \tilde{E}_x = E \cos \omega t, \quad \tilde{E}_y = E \sin \omega t$$

Squaring & adding, we get

$$\tilde{E}_x^2 + \tilde{E}_y^2 = E^2$$

This is the equation of circle.

So the polarization of em waves in which E-field vectors have equal magnitude & have phase difference 90° , is called circular polarization of EM waves.

In such a case, the locus of resultant $\vec{E}$ is a circle.

3) Elliptical Polarization:

(ii) When $E_1 \neq E_2$

$$\text{As } \vec{E}_x = E_1 \cos \omega t \Rightarrow \boxed{\frac{\vec{E}_x}{E_1} = \cos \omega t}$$

$$\vec{E}_y = -E_2 \sin \omega t \Rightarrow \boxed{\frac{\vec{E}_y}{E_2} = -\sin \omega t}$$

Squaring & adding the above two equations

$$\frac{E_x^2}{E_1^2} + \frac{E_y^2}{E_2^2} = 1$$

This is the equation of an ellipse

So, the polarization of em waves in which E-field vectors have different magnitude & a phase difference of 90° , then it is called elliptical polarization.

In this case, the locus of resultant E wave is an ellipse.

Brewster's Law of Polarization of light

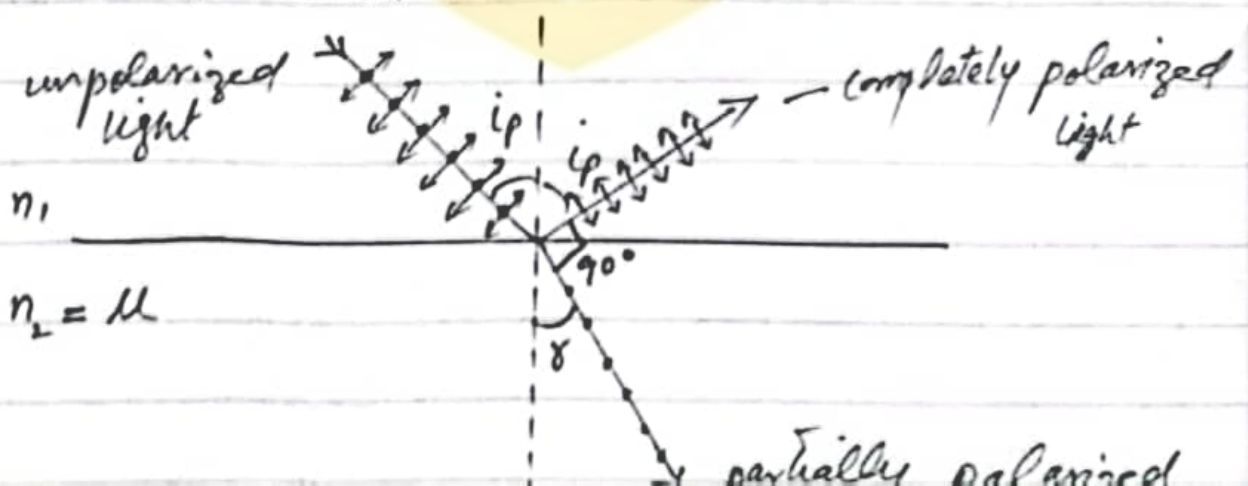
When an unpolarized light falls on the surface of a transparent material (water, glass etc) at some angle, some of the light is reflected & some of that is refracted.

If the angle b/w reflected & refracted ray is 90° , then reflected light is completely polarized, refracted light is partially polarized. The angle of incident with normal of the medium is called polarizing angle 'ip', the statement is called Brewster's law.

Mathematically

$$\mu = \tan i_p$$

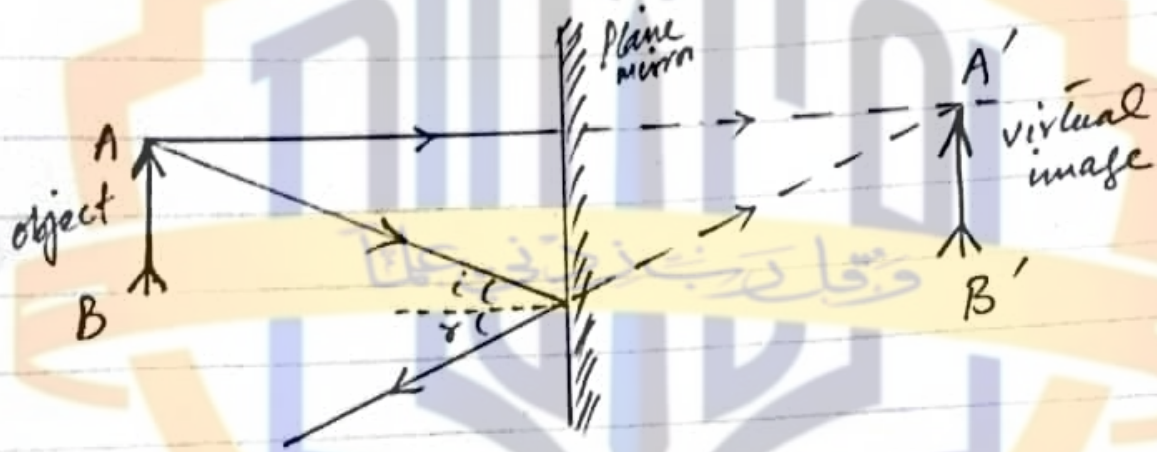
where " μ " is called the refractive index of 2nd medium w.r. to 1st medium.



Plane Mirror :-

A part of transparent ^{plane} sheet whose one side is polished & other side is reflecting (shining) is called. plane mirror. Usually back side is polished with silver or Aluminium.

Image formation by plane mirror is shown below as:



properties.

- i) plane mirror forms virtual & erect image.
- ii) size of image is equal to the size of the object.
- iii) Distance of image from mirror is equal to the distance of object from the mirror.
- iv) The image is laterally inverted.
e.g the image of R is Я.

Spherical Mirror:-

A part of hollow sphere whose one side is polished with Silver or Aluminium and other side is reflecting is called spherical mirror.

It is of two types:

i) Convex Mirror:- A spherical mirror whose outer side is polished and inner side is reflecting is called Convex mirror.

eg: side mirrors of vehicles.

Usually Convex mirror forms erect & virtual image.

ii) Concave Mirror:-

A spherical mirror whose inner side is polished & outer side is reflecting is called concave mirror.

eg: Mirror in hair dresser shop.

Usually Concave mirror forms real & inverted image.

Terminologies:-

Pole:- The central point of spherical mirror is called pole, denoted by "P".

Centre of curvature:-

The centre of sphere whose part is made mirror is called

centre of curvature, denoted by C .

Radius of curvature :-

The radius of sphere whose part is made sphere is called radius of curvature, denoted by " R ".

Focus point :-

The point near the spherical mirror where the parallel rays of light converge (meet) or appear to converge is called focus point, denoted by " F ".

Usually F is the mid point between C and P .

Focal length :-

The distance between pole of spherical mirror & focus point is called focal length, denoted by " f ". Usually $f = R/2$.

Principal axis :-

The line passing through centre of curvature & pole of mirror is called principal axis.

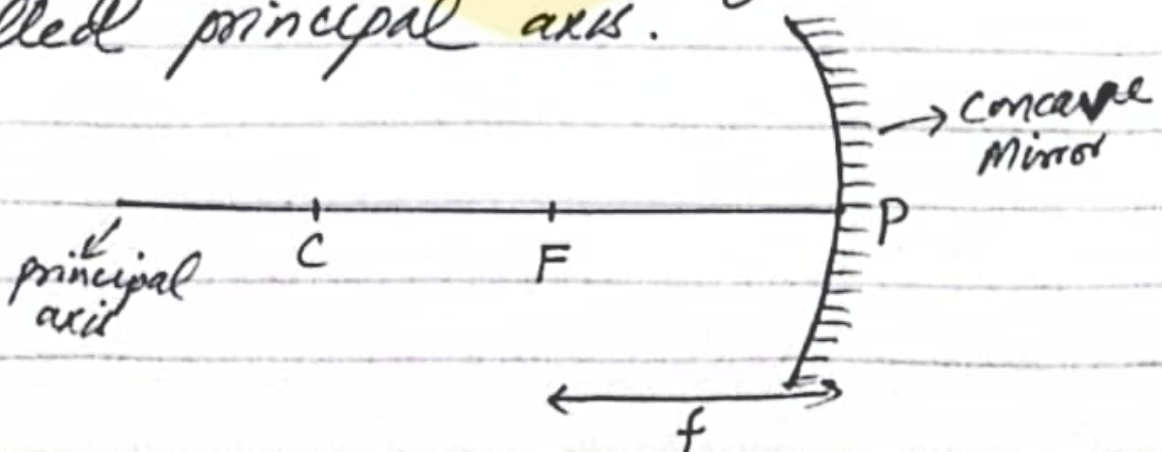
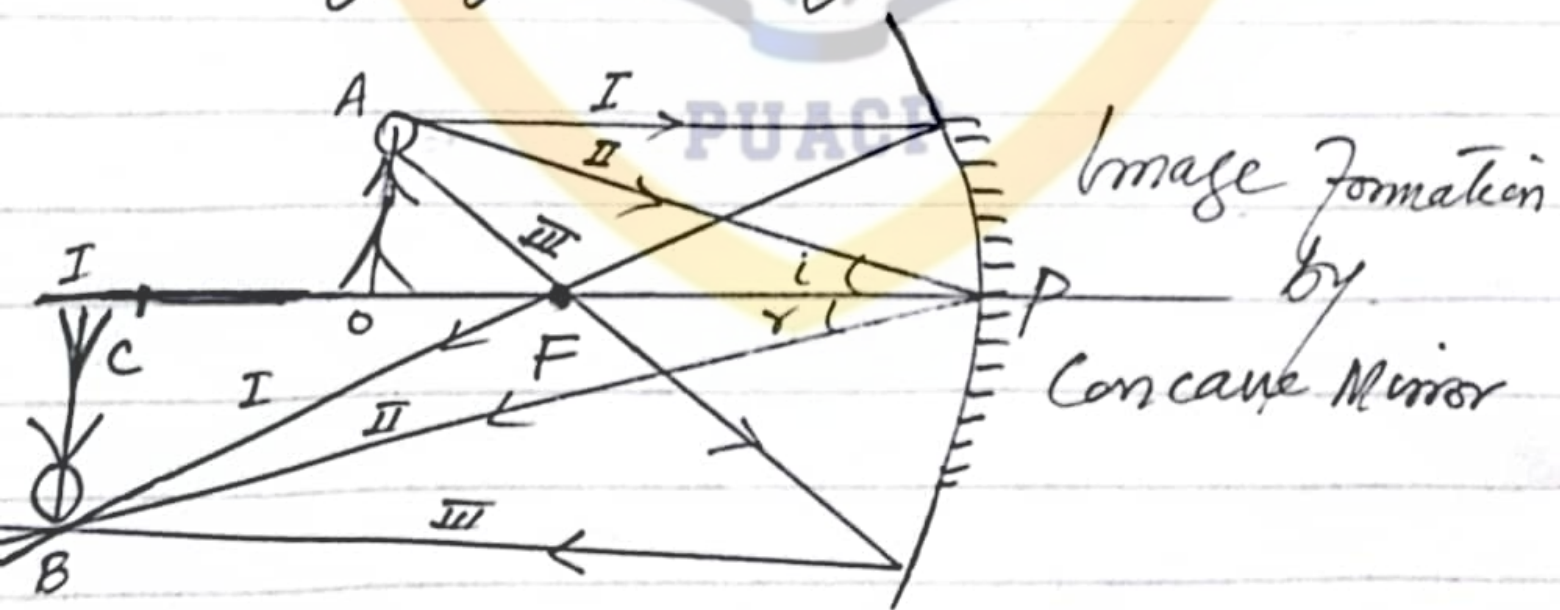


Image Formation: by ray diagram.

In spherical mirror, The image of an object can be formed by the intersection point of any two rays out of three rays. From the tip of object:

- i) A ray parallel to principal axis passes through focus point after reflection.
- ii) A ray passes through focus point becomes parallel to principal axis after reflection by mirror.
- iii) A ray incident at pole (P) is reflected back by satisfying the laws of reflection of light i.e. $\angle i = \angle r$.



Mirror's Equation:

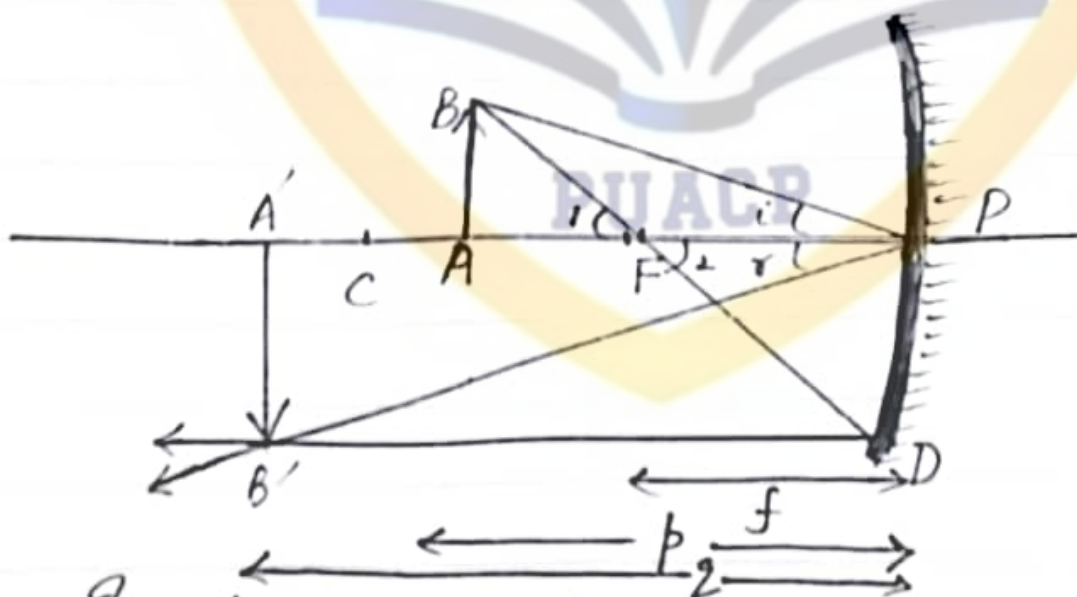
$$\frac{1}{f} = \frac{1}{p} + \frac{1}{q}$$

f = focal length of the object

p = distance of object from the concave mirror

q = distance of image from the concave mirror.

For the derivation of Mirror's equation, consider the following ray diagram for Concave mirror, in which $\overline{AB}$ is the object placed at a distance " p " from the mirror & forms an image $\overline{A'B'}$ at a distance " q " from the mirror.



In $\triangle ABP \leftrightarrow \triangle A'B'P$

$\angle APB \cong \angle A'PB'$

as $\angle i = \angle r$

$\angle BAP \cong \angle B'AP$ each $\cong 90^\circ$
 then $\angle ABP \cong \angle A'B'P$ each is complement of $\angle i$ & $\angle r$.

So $\triangle ABP \sim \triangle A'B'P$

$$\frac{AB}{A'B'} = \frac{AP}{A'P}$$

In similar Δ 's
 (ratio b/w corresponding
 sides is equal).

$$\boxed{\frac{AB}{A'B'} = \frac{p}{q}} \quad \text{--- (1)}$$

Now in $\triangle ABF \sim \triangle PDF$

$$\angle BAF \cong \angle DPF \quad (\text{each} \cong 90^\circ)$$

$$\angle 1 \cong \angle 2 \quad (\text{vertical angles})$$

$$\Rightarrow \angle FBA \cong \angle FDP \quad (\text{each is complement of } \angle 1 \text{ and } \angle 2)$$

So $\triangle ABF \sim \triangle PDF$

$$\frac{AB}{PD} = \frac{AF}{PF}$$

$$\frac{AB}{A'B'} = \frac{AP - PF}{PF}$$

$$\therefore PD = A'B'$$

$$\boxed{\frac{AB}{A'B'} = \frac{p-f}{f}} \quad \text{--- (2)}$$

Comparing eq (1) & (2)

$$\frac{p}{q} = \frac{p-f}{f}$$

$$pf = 2(p-f)$$

$$pf = 2p - 2f$$

Dividing by $p2f$ on BS

$$\frac{pf}{p2f} = \frac{2p}{p2f} - \frac{2f}{p2f}$$

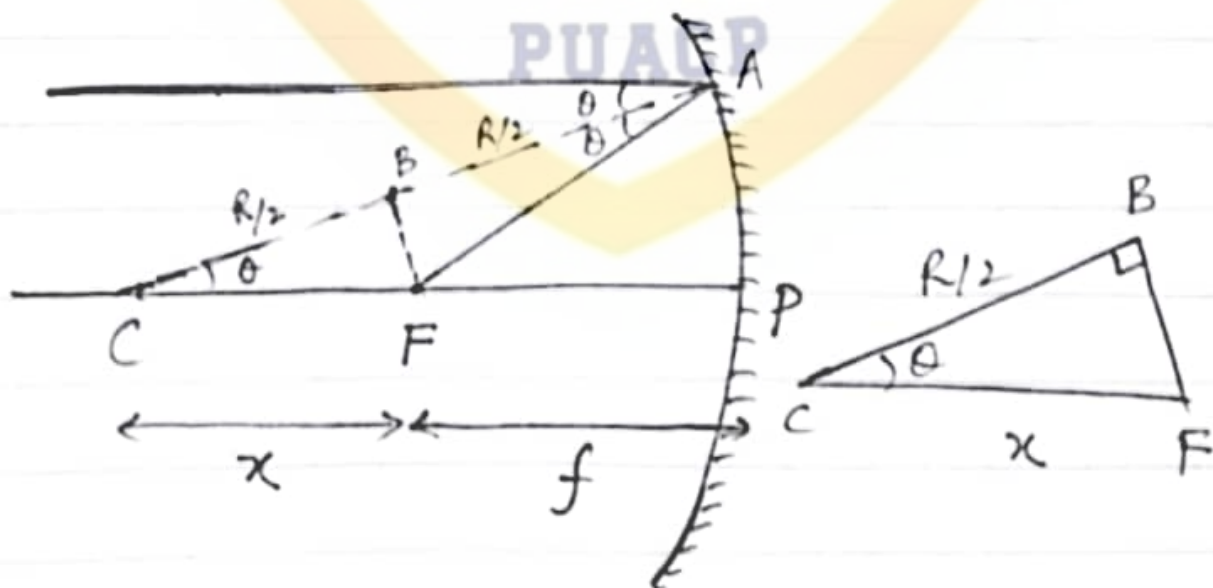
$$\frac{1}{2} = \frac{1}{f} - \frac{1}{p}$$

$$\frac{1}{p} + \frac{1}{2} = \frac{1}{f}$$

OR $\boxed{\frac{1}{f} = \frac{1}{p} + \frac{1}{2}}$

this is the required Mirror equation

Derivation for the relation of focal length of Spherical Mirror:-



(35)

In $\triangle BCF$

$$\cos \theta = \frac{R/2}{x} \Rightarrow \cos \theta = \frac{R}{2x}$$

$$x = \frac{R}{2 \cos \theta} = \frac{R}{2} \sec \theta$$

$$f = R - x = R - \frac{R}{2} \sec \theta.$$

If the mirror is of small aperture, then rays will be parallel and $\theta \ll 1$.

then $\sin \theta \approx \theta \approx 0$

$$\cos \theta \approx 1 \Rightarrow \sec \theta \approx 1$$

So, above relation becomes as:

$$f = R - \frac{R}{2}$$

$$f = R/2$$

The actual formula for focal length is

$f = R - \frac{R}{2} \sec \theta$ but usually $f = R/2$ in most of the cases.

Lens:-

A lens is an optical with perfect or approximate axial symmetry which transmit and refract light, converging or diverging the beam.

Lenses are made of glass or transparent

plastic. Elements which refract electromagnetic radiation outside the visual spectrum are also called lenses.

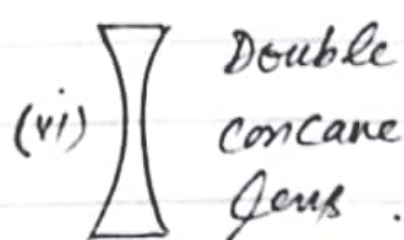
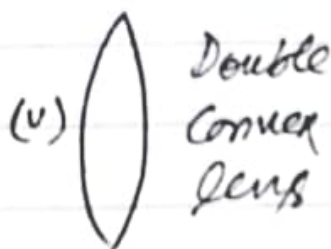
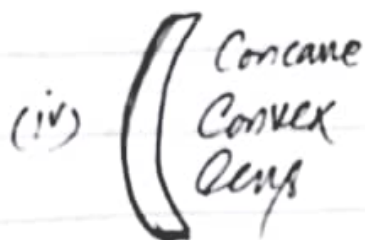
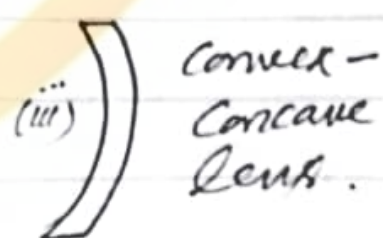
Thin lens :-

A lens whose thickness is negligible as compared to its focal length is called thin lens. These lenses have large focal length.

Thick lens :-

A lens whose thickness is not negligible & has small focal length is called thick lens.

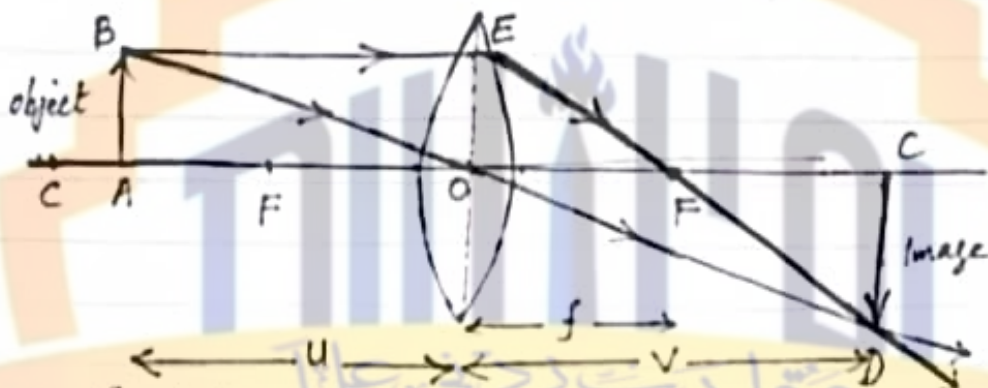
The lenses are of the following types.



(37)

Lens Equation:-

Consider the following ray diagram, in which AB is the object placed at a distance ' u ' from the double convex lens. Its image ED is formed behind the lens at a distance ' v ' from the lens. f be the focal length of the lens.



In $\triangle ABO \sim \triangle CDO$

$$\angle BAO \cong \angle DCO$$

each $\cong 90^\circ$

$$\angle AOB \cong \angle COD$$

vertical angles

$$\angle ABO \cong \angle CDO$$

each is complement of $\angle AOB$ & $\angle COD$

By the congruency of three angles

$$\triangle ABO \sim \triangle CDO$$

$$\frac{AB}{CD} = \frac{OA}{OC}$$

(ratio of corresponding sides are equal)

$$\frac{AB}{CD} = \frac{u}{v} \quad \text{--- (1)}$$

$\frac{1}{-u} + \frac{1}{v} = \frac{1}{f}$
 $\frac{1}{-u} + \frac{1}{v} = \frac{1}{f}$
 $\frac{1}{-u} + \frac{1}{v} = \frac{1}{f}$
 The required ray through spherical lens.
 the object's distance from the lens.

(38)

Now in $\triangle EOF \leftrightarrow \triangle CDF$

$$\angle EOF \cong \angle DCF \quad \text{each} \cong 90^\circ$$

$$\angle EFO \cong \angle CDF \quad \text{vertical angles}$$

$$\angle OFE \cong \angle CDF \quad \text{each is complement of } \angle EFO \text{ \& } \angle CDF$$

By the congruency of three angles

$$\triangle EOF \sim \triangle CDF$$

$$\frac{OE}{CD} = \frac{OF}{CF}$$

$$\frac{AB}{CD} = \frac{OF}{OC - OF}$$

$$\text{As } \overline{OE} \cong \overline{AB} \\ \overline{OC} - \overline{OF} = \overline{CF}$$

$$\frac{AB}{CD} = \frac{f}{v-f} \quad \text{--- (2)}$$

Comparing eq (1) & (2)

$$\frac{u}{v} = \frac{f}{v-f}$$

$$\Rightarrow u(v-f) = vf$$

$$\Rightarrow uv - uf = vf$$

$$\Rightarrow uv = vf + uf$$

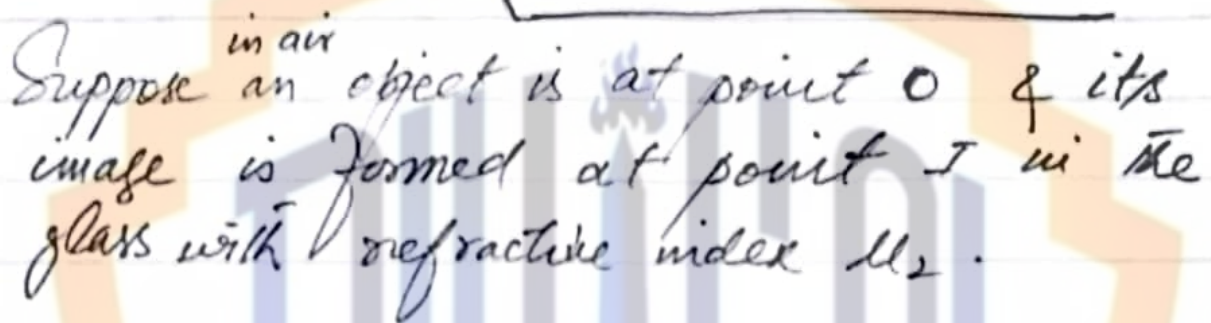
$$\Rightarrow \frac{uv}{uvf} = \frac{vf}{uvf} + \frac{uf}{uvf} \quad \text{Dividing by } uvf \text{ m.s.}$$

$$\Rightarrow \boxed{\frac{1}{f} = \frac{1}{u} + \frac{1}{v}}$$

which is the required Lens equation

$$\frac{\mu_1}{v} + \frac{\mu_2}{u} = \frac{\mu_2 - \mu_1}{R}$$

direct result
spherical
it's distance
from the focus


$$\mathcal{U}_1 \cap \mathcal{U}_2 = \mathcal{U}_1 \cap \mathcal{U}_2$$
$$\Rightarrow \mu_1 < \mu_2 = \mu_3 < \mu_4 \quad \text{--- (i)}$$
$$\angle i = \angle \alpha + \angle \gamma \quad \text{--- (ii)}$$
$$\angle Y = \angle x + \angle \beta$$
$$\Rightarrow \angle \gamma = \angle \gamma - \angle \beta \quad \text{--- (iii)}$$

putting values from eq (ii) & (iii) in eq (v)

$$\Rightarrow \mu_1(\angle \alpha + \angle \gamma) = \mu_2(\angle \gamma - \angle \beta)$$

using relation $l = rO \Rightarrow O = \frac{l}{r}$

$$\Rightarrow \mu_1 \left(\frac{AM}{OM} + \frac{AM}{CM} \right) = \mu_2 \left(\frac{AM}{CM} - \frac{AM}{IM} \right)$$

$$\mu_1 (AM) \left(\frac{1}{OM} + \frac{1}{CM} \right) = \mu_2 (AM) \left(\frac{1}{CM} - \frac{1}{IM} \right)$$

$$\mu_1 \left(\frac{1}{OM} + \frac{1}{CM} \right) = \mu_2 \left(\frac{1}{CM} - \frac{1}{IM} \right)$$

if M is very close to P, then above relation can be written as:

$$\mu_1 \left(\frac{1}{OP} + \frac{1}{CP} \right) = \mu_2 \left(\frac{1}{CP} - \frac{1}{IP} \right)$$

$$\frac{\mu_1}{OP} + \frac{\mu_1}{CP} = \frac{\mu_2}{CP} - \frac{\mu_2}{IP}$$

$$\frac{\mu_1}{-u} + \frac{\mu_1}{+R} = \frac{\mu_2}{+R} - \frac{\mu_2}{+v}$$

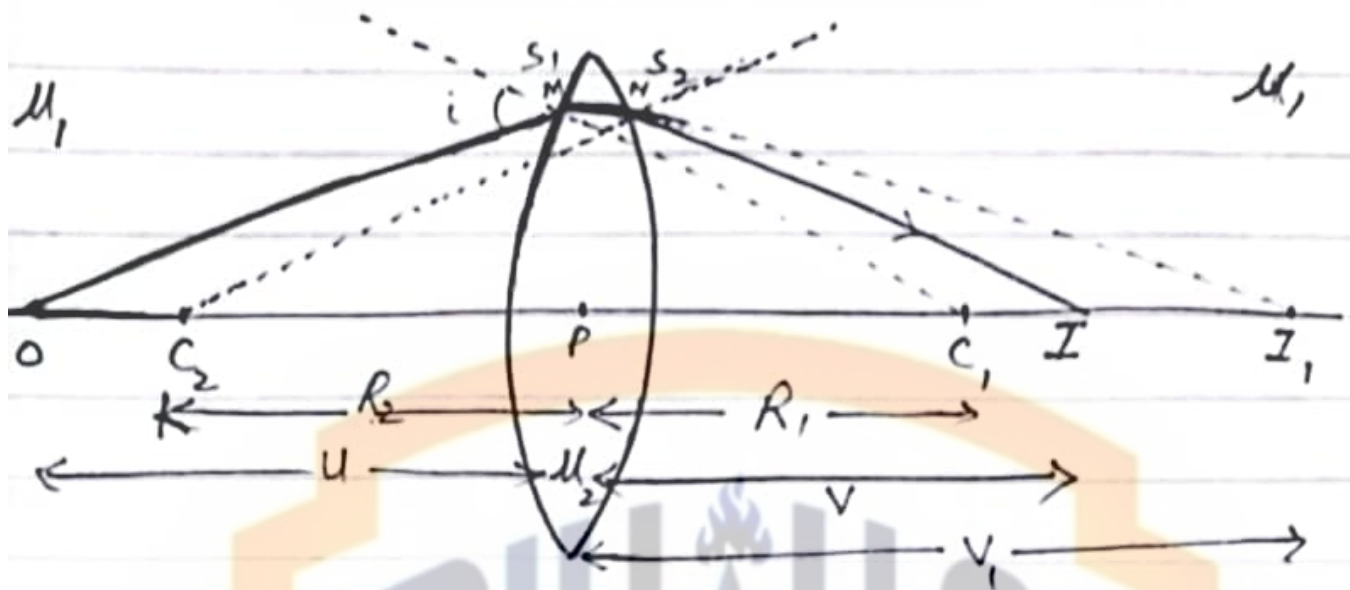
$$\frac{\mu_1}{-u} + \frac{\mu_2}{+v} = \frac{\mu_2}{+R} - \frac{\mu_1}{+R}$$

$$\boxed{\frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{1}{R} (\mu_2 - \mu_1)}$$

This is the required result for refraction of light through spherical surface & for real image.

Here u is the object's distance & v is the image distance from the boundary of two mediums.

Len Maker Formula.



Consider the object is placed at point O in front of a convex lens with left surface S_1 & right surface S_2 , left radius R_1 & right radius R_2 .

A ray of light is incident from O on S_1 at M & refract to MN. If it goes straight it forms image at point I_1 with distance v_1 from the lens. But the ray refracts again at point N & forms image at point I with image distance v .

Now for surface S_1 , from rare to denser medium, by the result of refraction through spherical surface.

$$-\frac{\mu_1}{u} + \frac{\mu_2}{v_1} = \frac{1}{R_1}(\mu_2 - \mu_1) \quad \text{--- (1)}$$

(42)

For surface R_2 . The refraction from denser to rare medium.

$$-\frac{\mu_2}{v_1} + \frac{\mu_1}{v} = \frac{1}{R_2} (\mu_1 - \mu_2)$$

$$-\left(\frac{\mu_2}{v_1} - \frac{\mu_1}{v}\right) = -\frac{1}{R_2} (\mu_2 - \mu_1)$$

$$\frac{\mu_2}{v_1} - \frac{\mu_1}{v} = \frac{1}{R_2} (\mu_2 - \mu_1) \quad \text{--- (2)}$$

Subtracting eq (2) from eq (1)

$$-\frac{\mu_1}{u} + \frac{\mu_2}{v_1} - \left(\frac{\mu_2}{v_1} - \frac{\mu_1}{v}\right) = (\mu_2 - \mu_1) \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\mu_1 \left(\frac{1}{v} - \frac{1}{u}\right) = (\mu_2 - \mu_1) \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{v} - \frac{1}{u} = \left(\frac{\mu_2}{\mu_1} - 1\right) \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\boxed{\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2}\right)}$$

$$\begin{cases} \mu_2 = \mu \\ \mu_1 = 1 \end{cases}$$

this is called the Lens Maker Equation or Lens maker formula.

$$\begin{cases} \text{if } u = \infty \\ \text{then } v = f \end{cases}$$

Compound Microscope:-

It is an optical device used to see magnified images of tiny objects. A good quality microscope can produce magnification of the order of 1000.

Construction:-

It consists of two convex lenses of short focal length, arranged coaxially at the ends of two sliding metal tubes.

i) Objective:- It is a convex lens of very short focal length " f_o " and a small aperture. It is positioned near the object to be examined.

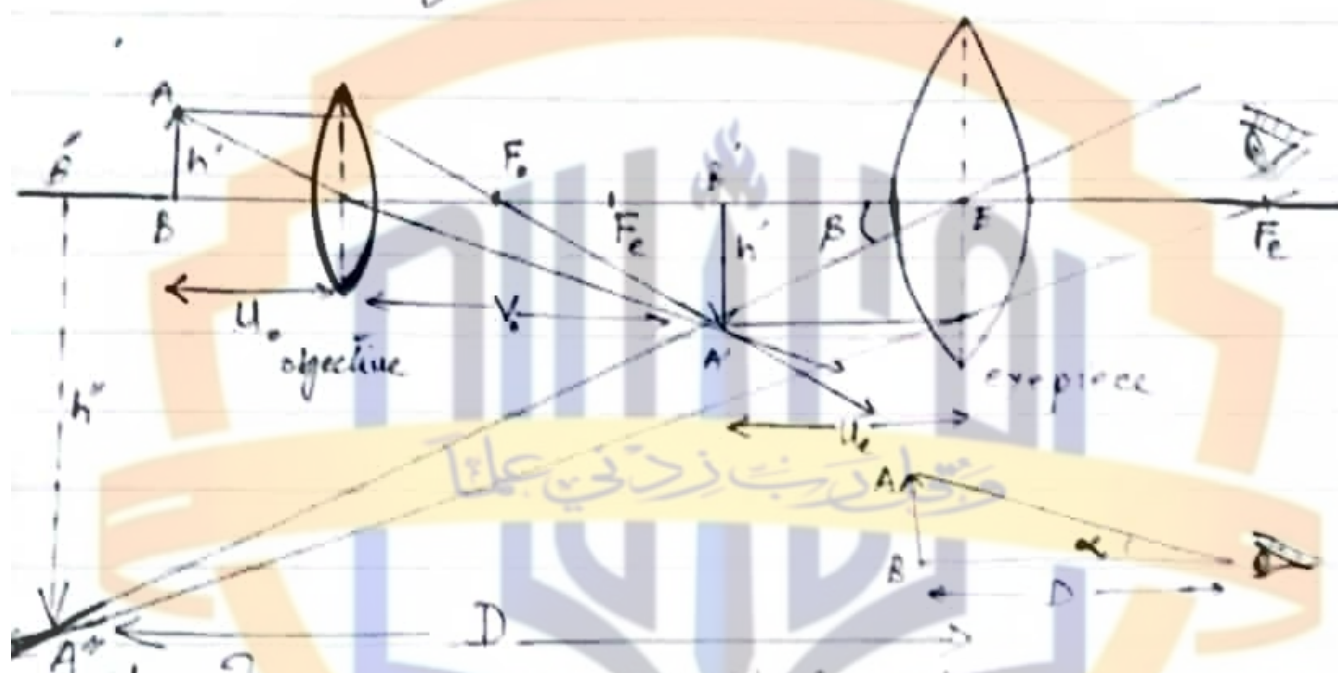
ii) Eye piece:- It is convex lens of larger focal length f_e than objective ($f_e > f_o$). It is positioned near the eye.

The distance between the two lenses can be varied by using rack & pinion arrangement.

Working:-

The object AB is placed at a distance u_o from the objective, objective forms it real image A'B' on the other side. The eyepiece is so adjusted that

$A'B'$ lies inside f_e . The image $A'B'$ of the object acts as an object for eyepiece. The eyepiece forms its final virtual, large & inverted image $A''B''$ at a distance D , as shown. The eyepiece acts as a simple microscope for $A'B'$.



Now for eyepiece as a simple microscope,

$$m_e = \frac{D}{U_e} = 1 + \frac{D}{f_e}$$

For objective

$$m_o = \frac{V_o}{U_o}$$

As we know that

$$m = m_o m_e$$

$$m = \frac{V_o}{U_o} \left(1 + \frac{D}{f_e} \right) \quad \text{--- (1)}$$

As the object is close to F_o , so $U_o \approx -f_o$.
Also, the image $A'B'$ is closer to eyepiece, so

$$v_o \approx L$$

So eq ① can be written as:

$$m = -\frac{L}{f_o} \left(1 + \frac{D}{f_e}\right)$$

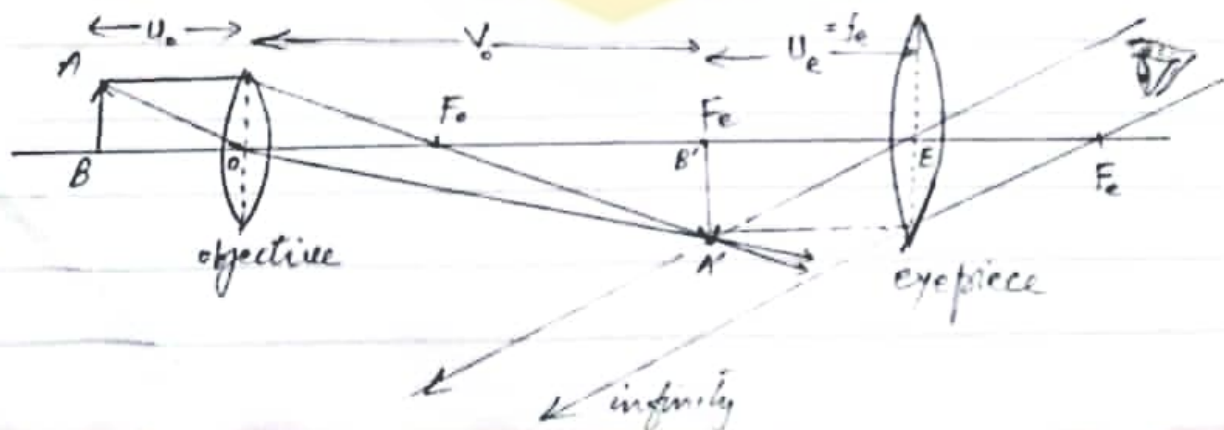
This is the relation for magnification, when the final image is formed at least distance of distinct vision.

Now we consider the case when the final image is formed at infinity.
In this case image $A'B'$ is on the focus F_e .
i.e. $u_e = f_e$, then $A'B'$ is formed at infinity.
So $m_e = D/f_e$, $m_o = \frac{L}{f_o}$

$$\Rightarrow m = m_o \cdot m_e$$

$$m = -\frac{L}{f_o} \times \frac{D}{f_e}$$

Magnifying power of compound microscope is large when both f_o and f_e are short / small.



Telescope:-

It is an optical device which enables us to see distant objects clearly. It provides angular magnification of the distant objects.

Refracting Telescopes is of two types.

- i) Astronomical telescope:- It is used to see the heavenly objects like the sun, stars, planets etc. The final image is inverted one which is immaterial in case of heavenly bodies because of their round shape.
- ii) Terrestrial Telescope:- It is used to see distant objects on the surface of the earth. The final image formed is erect one. This is an essential condition of viewing the objects on earth's surface correctly.

(i) Astronomical Telescope:-

It consists of two convex lenses mounted coaxially at the ends of two sliding tubes.

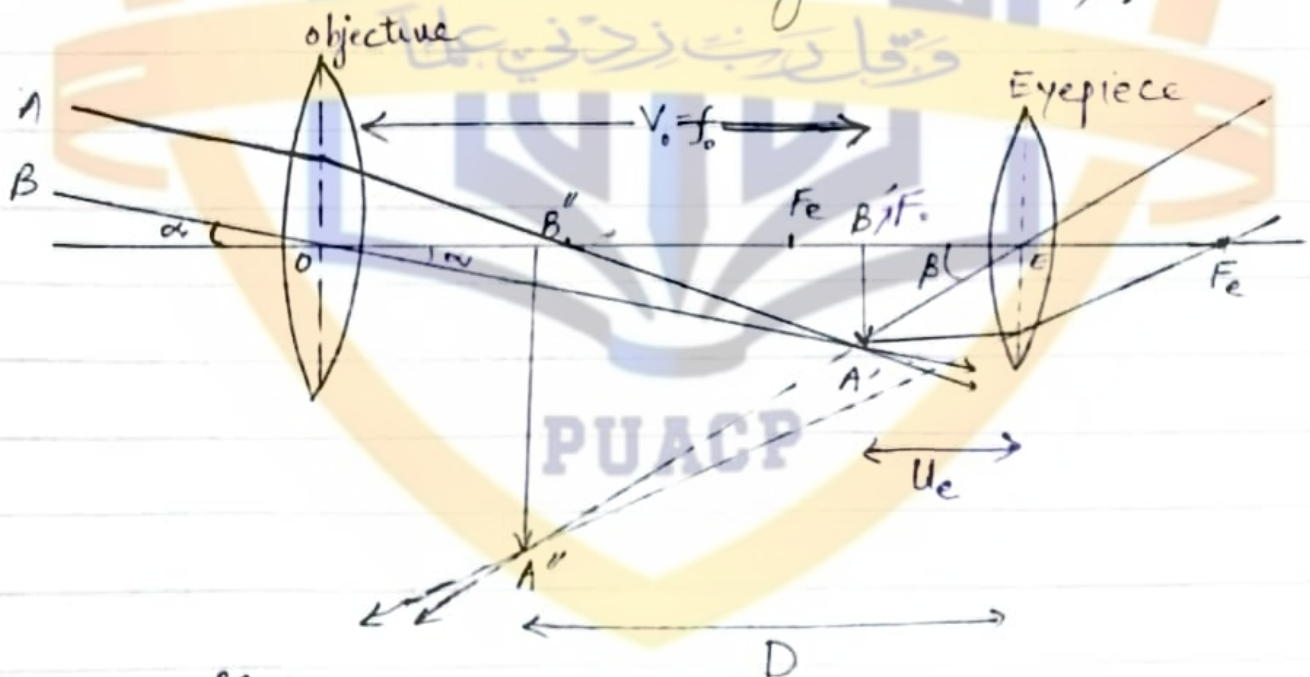
- a) The objective with large focal length & large aperture to gather sufficient light from the distant object. It forms bright image of distant object.
- b) The eyepiece with small focal length &

Small aperture so that the whole light may enter the eye for distant vision.

Working:-

(i) When the final image is formed at the least distance of distinct vision.

The parallel rays from the distant object AB fall on the objective at an angle α with principal axis. The objective forms a real, inverted & diminished image $A'B'$ inside the focus point of eyepiece. So that eyepiece forms a virtual, magnified & inverted final image $A''B''$ at a distance " D " from the eyepiece.



Magnifying power:-

The magnifying power of Telescope is the ratio of angle subtended by the final image at the eye " β " to the angle subtended by far object at the eye " α ".

$$m = \frac{\beta}{\alpha}$$

$$m = \frac{\tan \beta}{\tan \alpha}$$

if α, β are small.

$$m = \frac{A'B'/OB'}{A'B'/EB'} = \frac{EB'}{OB'}$$

Now $OB' = +f_o$, $BE = -u_e$

$$m = -\frac{f_o}{u_e}$$

For eyepiece

$$u = +u_e, v = -D$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$-\frac{1}{D} + \frac{1}{-u_e} = \frac{1}{f_e}$$

$$-\frac{1}{u_e} = \frac{1}{f_e} + \frac{1}{D} = \frac{1}{f_e} \left(1 + \frac{f_e}{D} \right)$$

So $m = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$

For large magnification,

$f_o \gg f_e$, Negative sign for inverted image

(ii) When the final image formed at Infinity:-

In this case the image $A'B'$ of far object AB is formed exactly at focus point F_e of eye piece which forms it virtual inverted & magnified image at infinity. As we know that the magnification of telescope is given as:

$$m = \beta/\alpha$$

OR

$$m = \frac{\tan \beta}{\tan \alpha}$$

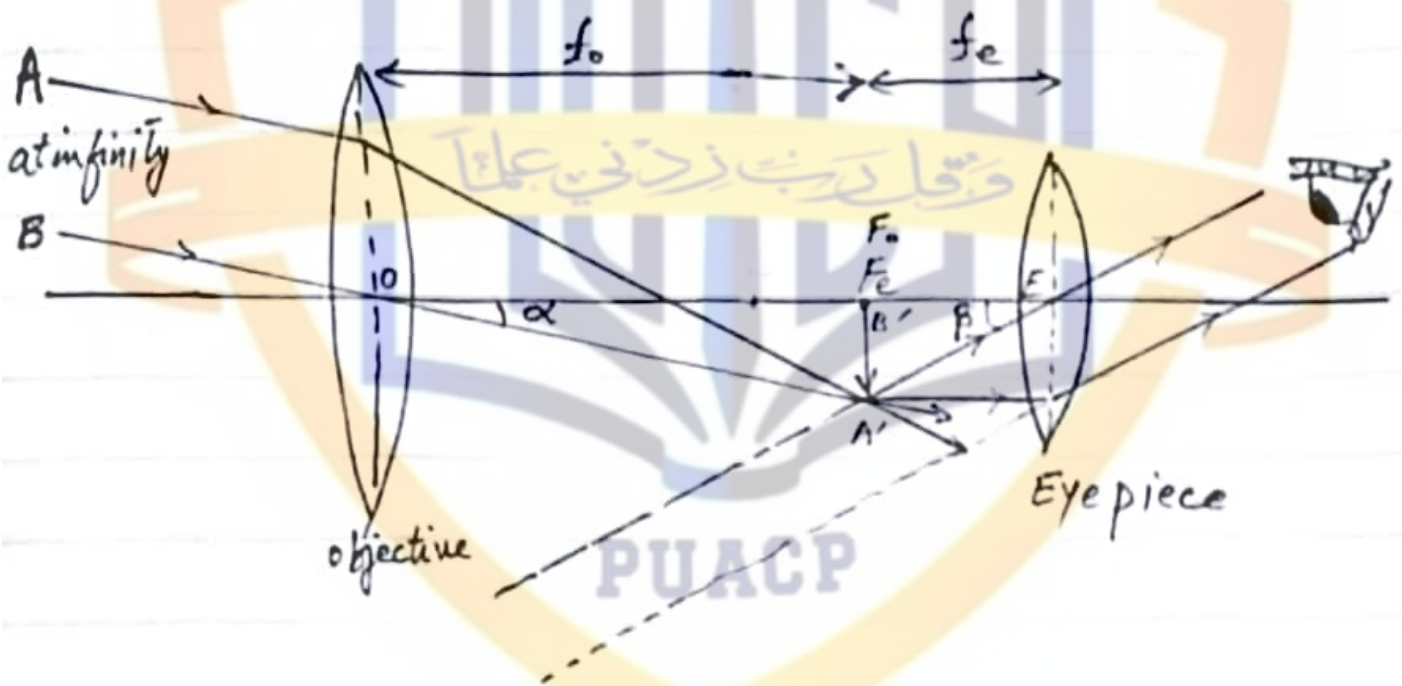
$$m = \frac{A'B'/BE}{A'B'/OB'} = \frac{OB'}{EB'}$$

$$OB' = +f_o, \quad B'E = -f_e$$

(for α, β are small)

$$\text{So } m = -f_o/f_e$$

m is large if $f_o \gg f_e$



(ii) Terrestrial Telescope:-

There is an additional convex lens between objective and eyepiece in this type of telescope to see the distant object clearly.

The objective forms a real & inverted image $A'B'$ of the object AB in its focal point. The erecting lens forms a real & inverted image $A''B''$ of the same size at its focus point. This image is erect with respect to far object. The eyepiece is so adjusted that $A''B''$ is at the focus point F_e of eyepiece. Finally the eyepiece forms a virtual, erect and a magnified image at infinity.

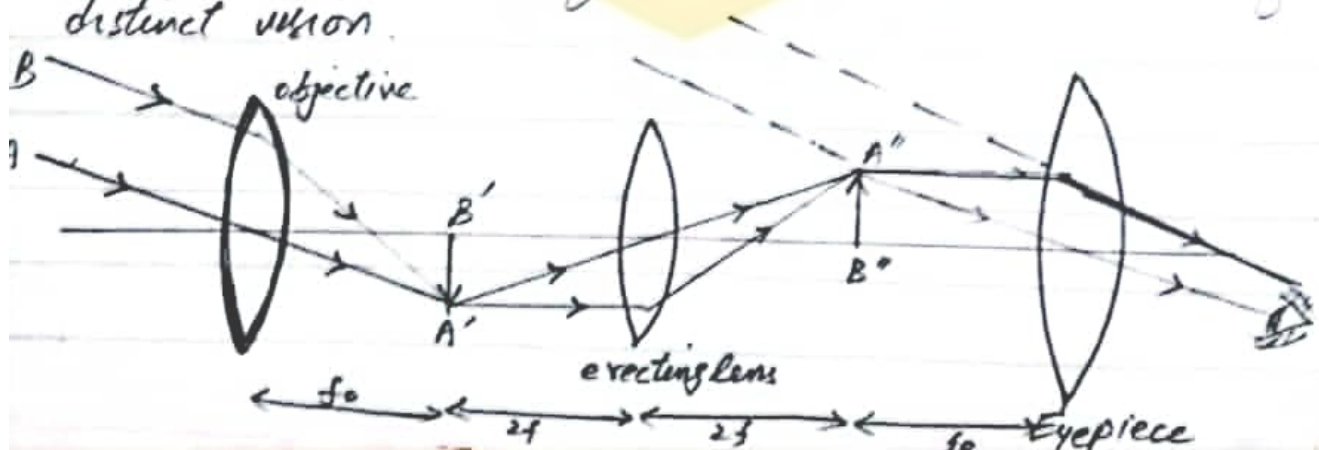
As the erecting lens does not cause any magnification, the angular magnification of the terrestrial telescope is same as that of the astronomical telescope i.e.

$$m = -f_o / f_e$$

When the image is formed at infinity,

$$m = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D}\right)$$

When the image is formed at least distance of distinct vision.



Drawbacks:-

i) The length of Terrestrial telescope is much larger than the astronomical telescope.

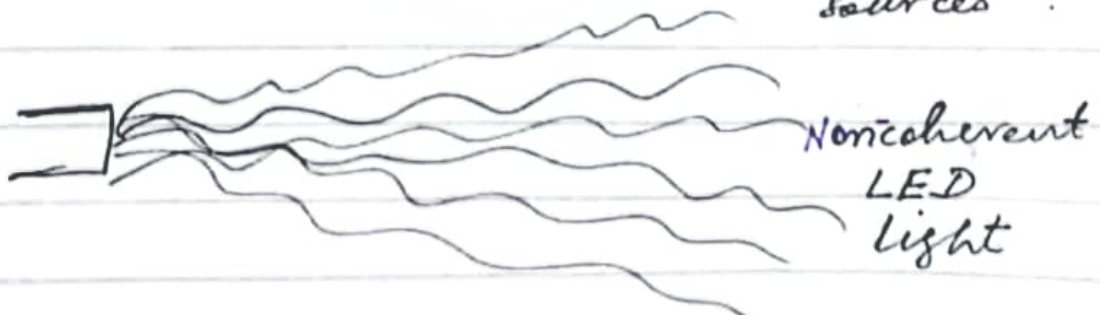
In normal adjustment, length of this telescope is $f_o + 4f + f_e$ (f = focal length of erecting lens)

ii) Due to extra reflection at the surface of erecting lens, the intensity of the final image decreases.

Coherent Sources:-

The two sources which produce waves having the same frequency, equal or comparable amplitude and zero or constant phase difference are called coherent sources.

eg: light from a Laser is emitted from the coherent sources.



Principle of Superposition:-

According to this principle, "The resultant displacement caused by two or more waves travelling in the same space is the algebraic sum of the displacements of the individual waves".

If individual waves have displacements $y_1, y_2, y_3, y_4, \dots$ and so on then the resultant wave will have the displacement y , given as:

$$y = y_1 + y_2 + y_3 + y_4 + \dots + y_n$$

Interference of light waves:-

The overlapping of two coherent light waves travelling in the same direction is called interference of waves.

The Interference pattern consists of maxima and minima.

Interference is of two types:-

- i) Constructive Interference.
- ii) Destructive Interference.

(i) Constructive Interference.

The overlapping of waves in which the resultant wave has the amplitude equal to the sum of amplitudes of individual waves, is called constructive interference. i.e.

$$Y = Y_1 + Y_2 \quad \text{or} \quad A = A_1 + A_2$$

(a) The path difference b/w two waves is given as

$$\boxed{\Delta x = n\lambda} \quad \text{where } n = 0, 1, 2, 3, 4, \dots$$

(b) The phase difference b/w two waves is given as:

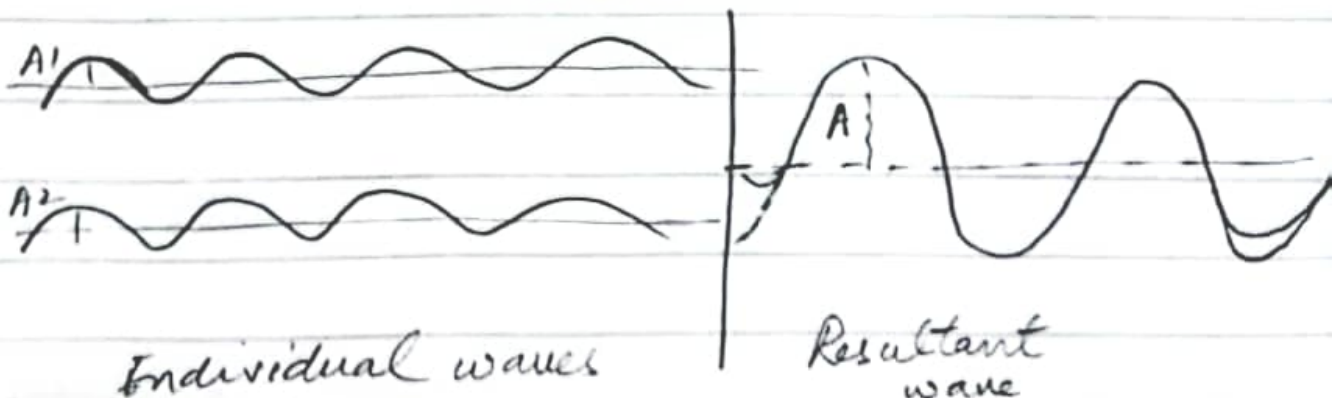
$$\boxed{\phi = 2n\pi} \quad \text{where } n = 0, 1, 2, 3, \dots$$

$$\Rightarrow \cos \phi = 1 \quad (\text{Maximum})$$

(c) The resultant wave has intensity I , given as:

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi = I_1 + I_2 + 2\sqrt{I_1 I_2}$$

$$\Rightarrow A^2 = A_1^2 + A_2^2 + 2A_1 A_2 \cos \phi = A_1^2 + A_2^2 + 2A_1 A_2$$



(ii) Destructive Interference :-

The overlapping of waves in which, the resultant wave has the amplitude equal to the difference of the amplitudes of the individual waves.

$$\text{i.e. } Y = Y_1 - Y_2 \quad \text{or} \quad Y = Y_2 - Y_1$$

$$\text{OR } A = A_1 - A_2 \quad \text{or} \quad A = A_2 - A_1$$

(a) The path difference between two waves is given by:

$$\Delta x = (n + \frac{1}{2}) \lambda \quad \text{where } n = 0, 1, 2, 3, \dots$$

(b) The phase difference between two waves is given by;

$$\phi = (2n - 1) \pi \quad \text{where } n = 0, 1, 2, 3, \dots$$

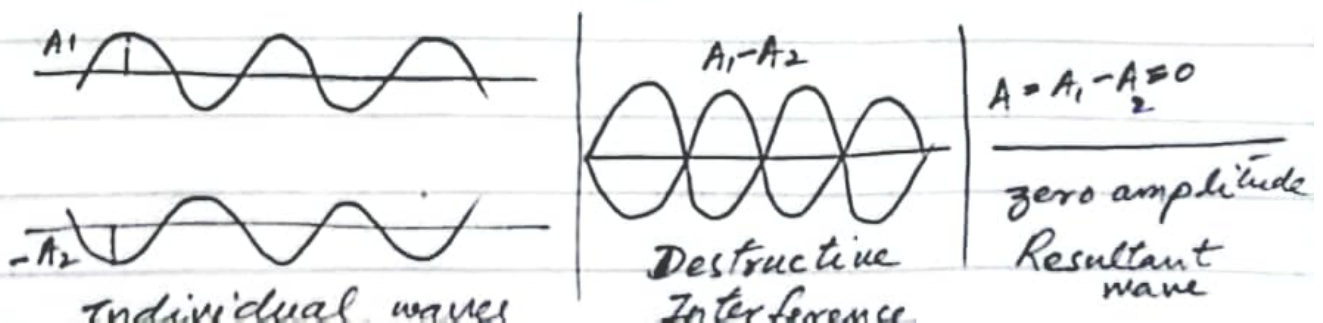
$$\Rightarrow \cos \phi = -1 \quad (\text{minimum})$$

(c) The resultant wave has the intensity I , given as:

$$I = \frac{I_1}{1} + \frac{I_2}{2} - 2\sqrt{I_1 I_2} \quad (\cos \phi = -1).$$

OR

$$A^2 = A_1^2 + A_2^2 - 2A_1 A_2$$



Refractive Index (μ) :-

The refractive index of a material is defined as the ratio of speed of light in vacuum to the speed of light in the medium.

It is denoted by " μ ". It has no unit as it is the ratio between similar quantities.

$$\text{Refractive Index} = \mu$$

$$= \frac{\text{Speed of light in vacuum}}{\text{Speed of light in the medium}}$$

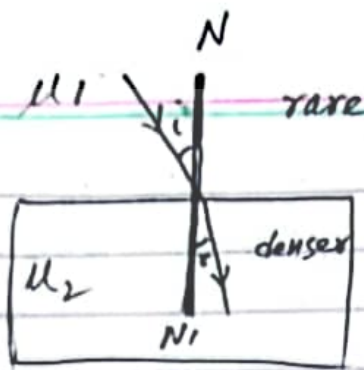
$$= \frac{c}{v}$$

The refractive Index depends on the optical density of the medium which means the value of resistance in passing light through the medium due to the arrangement of atoms in the medium. It is similar to mass density but a quite different term.

Most of the object has greater optical density as well as greater mass density but some material have less optical density even having greater mass density.

eg oil has greater optical density than that of water but mass density of water is greater than oil.

A material having greater optical density has large refractive index and causes greater refraction (bending) of light ray through it.



By Snell's Law

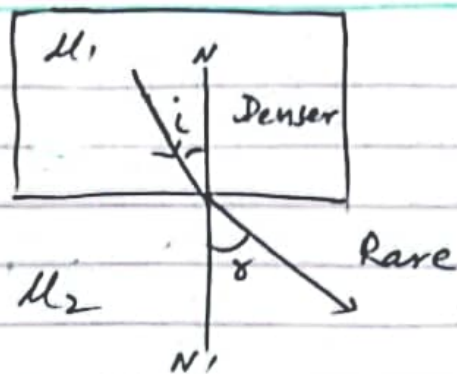
$$\mu_1 \sin i = \mu_2 \sin r$$

$$\frac{\mu_1}{\mu_2} = \frac{\sin r}{\sin i}$$

$$\frac{\mu_2}{\mu_1} = \frac{\sin i}{\sin r}$$

For air $\mu_1 = 1, \mu_2 = \mu$

$$\Rightarrow \mu = \frac{\sin i}{\sin r}$$



$$\frac{\mu_1}{\mu_2} = \frac{\sin r}{\sin i}$$

$$\Rightarrow \frac{\mu}{1} = \frac{\sin r}{\sin i}$$

$$\mu_1 = \mu$$

$$\mu_2 = 1$$

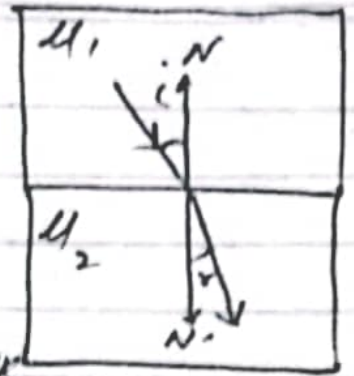
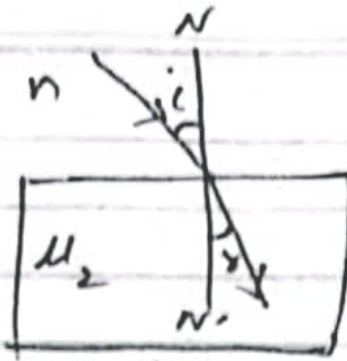
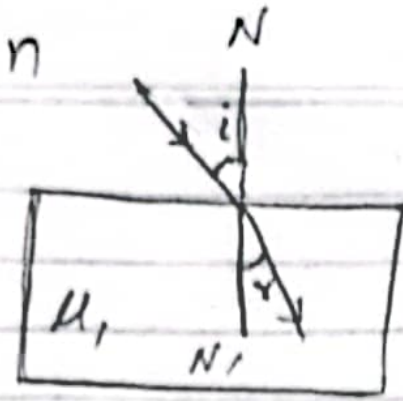
$$\Rightarrow \mu = \frac{\sin r}{\sin i}$$

After refraction, the wave has same frequency but different speed, wavelength and direction.

If ray of light enters from rare into denser medium, then it bends towards the normal of the two mediums i.e. $i > r$.

If ray of light enters from denser into rare medium, then it bends away from the normal i.e. $i < r$.

If we have to find the refractive index of one medium with respect to other medium rather with respect to air. Then let one medium has refractive index μ , and other medium has μ_2 , air has n , then



By definition of Refractive Index
 $\mu_1 = \frac{c}{v_1}$ and $\mu_2 = \frac{c}{v_2}$

By dividing the two relations

$$\frac{\mu_1}{\mu_2} = \frac{c/v_1}{c/v_2}$$

$$\frac{\mu_1}{\mu_2} = \frac{1/v_1}{1/v_2}$$

$$\boxed{\frac{\mu_1}{\mu_2} = \frac{v_2}{v_1}}$$

$$\text{or } \boxed{\frac{\mu_2}{\mu_1} = \frac{v_1}{v_2}}$$

The other formulae of Refractive Index are.

$$\mu = \frac{\text{Real depth of a liquid}}{\text{Apparent depth of the liquid}}$$

$$\mu = \frac{1}{\sin C}$$

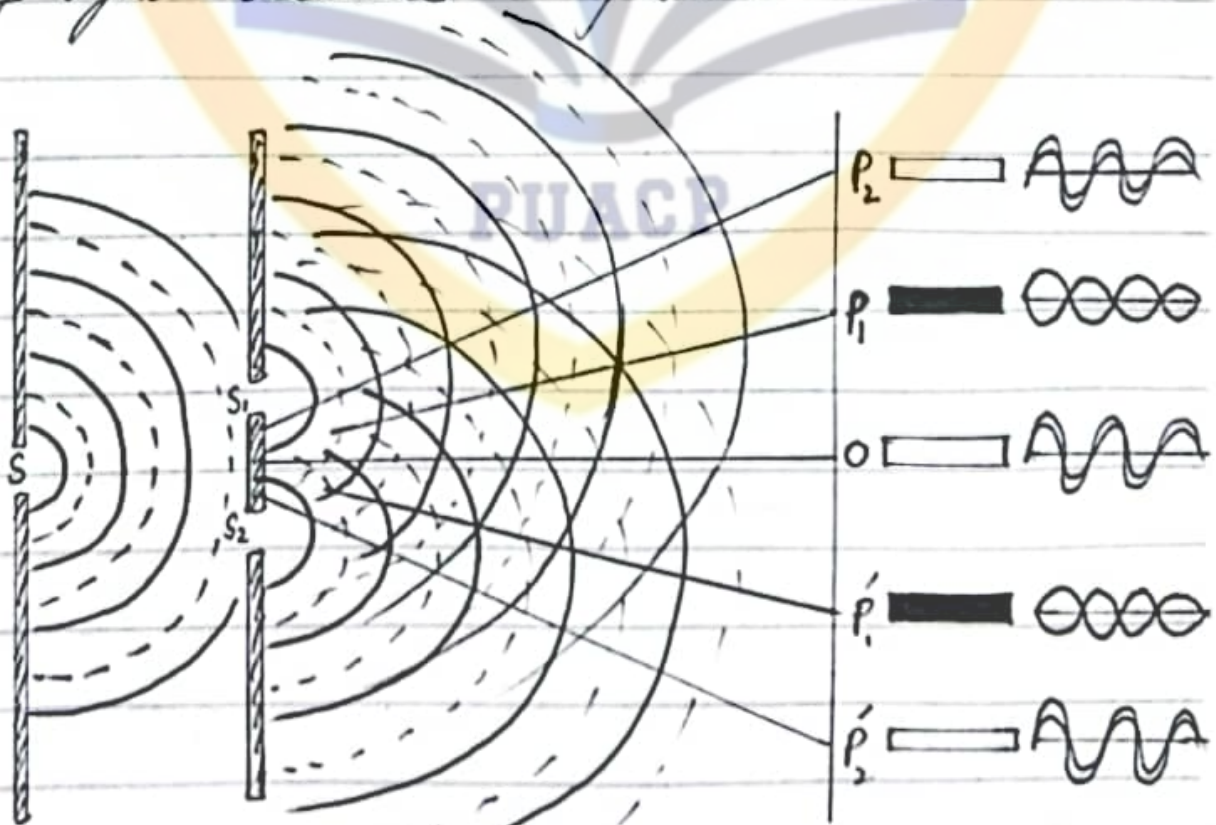
where
 $C = \text{Critical angle.}$

Young's Double Slit Experiment:-

In 1801, Thomas Young was the first person, who demonstrated experimentally the interference of light.

In his experiment he placed a source " S " of monochromatic light (a sodium vapour lamp). At a distance of 10 cm from S , he placed two narrow slits S_1 and S_2 , separated by 2 mm, each having width 0.3 mm. A screen is placed 2 m away from S_1 & S_2 where dark and bright fringes are formed after interference of two light beams.

When one of the slits S_1 or S_2 is closed then fringes disappear and the intensity of light becomes uniform.



In figures the solid arcs represent crests while dotted arcs represent troughs.

At point O, P, P' constructive interference occur & bright fringes^{are} formed (where $I \propto A^2$).

At point P, & P' destructive interference occur & dark fringes are formed.

Let y_1 and y_2 be the displacement of two waves having phase difference ϕ .

ie $y_1 = a_1 \sin \omega t$

$y_2 = a_2 \sin(\omega t + \phi)$ (a_1, a_2 are amplitudes)

By the principle of superposition Resultant wave has displacement y given as

$$y = y_1 + y_2$$

$$y = a_1 \sin \omega t + a_2 \sin(\omega t + \phi)$$

$$y = a_1 \sin \omega t + a_2 \sin \omega t \cos \phi + a_2 \cos \omega t \sin \phi$$

$$y = (a_1 + a_2 \cos \phi) \sin \omega t + a_2 \sin \phi \cos \omega t \quad \text{--- (1)}$$

Let $a_1 + a_2 \cos \phi = A \cos \theta$ --- (i)

$a_2 \sin \phi = A \sin \theta$ --- (ii)

Then eq (1) becomes as:

$$y = A \cos \theta \sin \omega t + A \sin \theta \cos \omega t$$

$$y = A \sin(\omega t + \theta) \quad \text{--- (2)}$$

It is also harmonic wave with amplitude A & have phase difference θ with first

wave.

Intensity of Resultant wave:-

$$As \quad a_1 + a_2 \cos \phi = A \cos \theta$$

$$a_2 \sin \phi = A \sin \theta.$$

Squaring and adding these two equations.

$$a_1^2 + a_2^2 \cos^2 \phi + 2a_1 a_2 \cos \phi + a_2^2 \sin^2 \phi = A^2 \cos^2 \theta + A^2 \sin^2 \theta.$$

$$\Rightarrow A^2 (\cos^2 \theta + \sin^2 \theta) = a_1^2 + a_2^2 (\cos^2 \phi + \sin^2 \phi) + 2a_1 a_2 \cos \phi.$$

$$\Rightarrow A^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos \phi \quad \text{--- (3)}$$

As the intensity of wave is directly proportional to the square of amplitude, i.e.

$$I \propto A^2$$

$$\Rightarrow I = kA^2$$

So we can write $I_1 = k a_1^2$, $I_2 = k a_2^2$

From eq (3)

$$kA^2 = k a_1^2 + k a_2^2 + 2\sqrt{k a_1} \sqrt{k a_2} \cos \phi.$$

$$\Rightarrow I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi \quad \text{--- (4)}$$

This is the required relation for intensity I of resultant wave, Here I_1, I_2 are the intensities of individual terms and $\cos \phi$ is the interference term.

i) Constructive Interference:-

As we know that

$$\frac{\Delta x}{\lambda} = \frac{\Delta \phi}{2\pi}$$

$$\left\{ \begin{array}{l} \text{P.D} = n\lambda \\ \Delta x = n\lambda \times \frac{2\pi}{2\pi} \\ \frac{\Delta x}{\lambda} = \frac{2n\pi}{2\pi} \\ \frac{\Delta x}{\lambda} = \frac{\Delta \phi}{2\pi} \end{array} \right.$$

So in this case $\Delta \phi = 0, 2\pi, 4\pi, \dots$

$$\text{or } \boxed{\Delta \phi = 2n\pi} \quad \text{where } n = 0, 1, 2, \dots$$

and

$$\Delta x = 0, \lambda, 2\lambda, 3\lambda, \dots$$

$$\text{or } \boxed{\Delta x = n\lambda} \quad \text{where } n = 0, 1, 2, 3, \dots$$

using eq (4)

$$I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2} \quad (1)$$

$$\text{or } \Rightarrow \frac{I_{\max}}{I_1 + I_2} > 1$$

$$\text{Also } A^2 = a_1^2 + a_2^2 + 2a_1 a_2$$

$$\Rightarrow A^2 = (a_1 + a_2)^2$$

ii) Destructive Interference.

As we know that

$$\frac{\Delta x}{\lambda} = \frac{\Delta \phi}{2\pi}$$

In this case

$$\Delta \phi = \pi, 3\pi, 5\pi, \dots$$

$$\boxed{\Delta \phi = (2n+1)\pi} \quad \text{where } n = 0, 1, 2, 3, \dots$$

and $\frac{\Delta x}{\lambda} = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \dots$

$$\Delta x = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \frac{7\lambda}{2}, \dots \quad \text{where}$$

$$\Delta x = (2n+1) \frac{\lambda}{2} \quad n = 0, 1, 2, 3, \dots$$

$$\boxed{\Delta x = (2n+1) \frac{\lambda}{2}}$$

using eq (4)

$$I_{\min} = I_1 + I_2 + 2\sqrt{I_1 I_2}(-1)$$

$$I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$

$$I_{\min} < I_1 + I_2$$

Also $A^2 = a_1^2 + a_2^2 - 2a_1 a_2$

$$A^2 = (a_1 - a_2)^2$$

Now average intensity is given as:

$$I_{av} = \frac{I_{\max} + I_{\min}}{2}$$

$$I_{av} = \frac{(a_1 + a_2)^2 + (a_1 - a_2)^2}{2}$$

$$\boxed{I_{av} = a_1^2 + a_2^2}$$

If there is no interference b/w two waves
then

$$I = I_1 + I_2$$

$$I = a^2 + a^2$$

Comparison between $I_{\max}$ & $I_{\min}$

As $I_1 = k a_1^2$

$I_2 = k a_2^2$

Then $I_1 / I_2 = a_1^2 / a_2^2$

Similarly

$$\frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2}$$

$$\frac{I_{\max}}{I_{\min}} = \frac{\left(\frac{a_1}{a_2} + 1\right)^2}{\left(\frac{a_1}{a_2} - 1\right)^2}$$

$$\frac{I_{\max}}{I_{\min}} = \frac{(\gamma + 1)^2}{(\gamma - 1)^2}$$

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{\gamma + 1}{\gamma - 1}\right)^2$$

where $\gamma = \frac{a_1}{a_2} = \frac{\sqrt{I_1}}{\sqrt{I_2}}$ = amplitude ratio of the two waves.

Fringe Spacing :-

For maxima

$$m\lambda = d \sin \theta$$

$$m\lambda = d \tan \theta$$

$$m\lambda = d \frac{y_m}{L}$$

$$y_m = \frac{m\lambda L}{d}$$

$$y_1 = \frac{\lambda L}{d}, \quad y_2 = \frac{2\lambda L}{d}, \quad y_3 = \frac{3\lambda L}{d}$$

$$\Delta y = y_2 - y_1$$

$$\Delta y = \frac{2\lambda L}{d} - \frac{\lambda L}{d}$$

$$\boxed{\Delta y = \frac{\lambda L}{d}}$$

which is the required width between two consecutive bright fringes. This value is same for two consecutive dark fringes too.

For Minima :-

$$(m + \frac{1}{2})\lambda = d \tan \theta$$

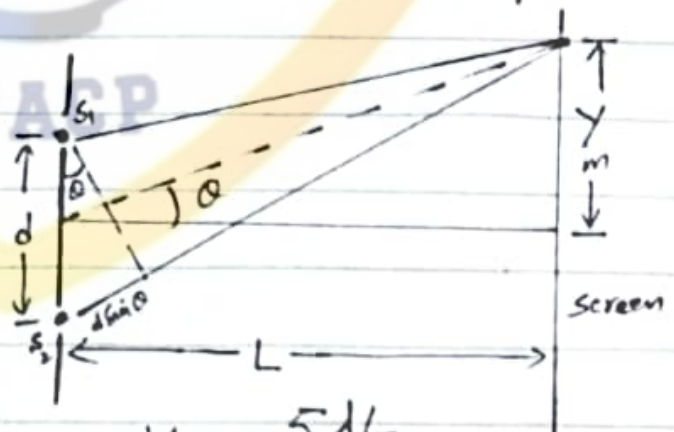
$$(m + \frac{1}{2})\lambda = d \frac{y_m}{L}$$

$$y_m = \frac{(m + \frac{1}{2})\lambda L}{d}$$

$$y_1 = \frac{\lambda L}{2d}, \quad y_2 = \frac{3\lambda L}{2d}, \quad y_3 = \frac{5\lambda L}{2d}$$

$$\Delta y = y_2 - y_1$$

$$\Delta y = \frac{3\lambda L}{2d} - \frac{\lambda L}{2d} = \frac{2\lambda L}{2d} = \frac{\lambda L}{d} \Rightarrow \boxed{\Delta y = \frac{\lambda L}{d}}$$



Interference from Thin Film :-

Thin film like soap bubbles or drops of oil spread on the surface of water show brilliant colours when exposed to light.

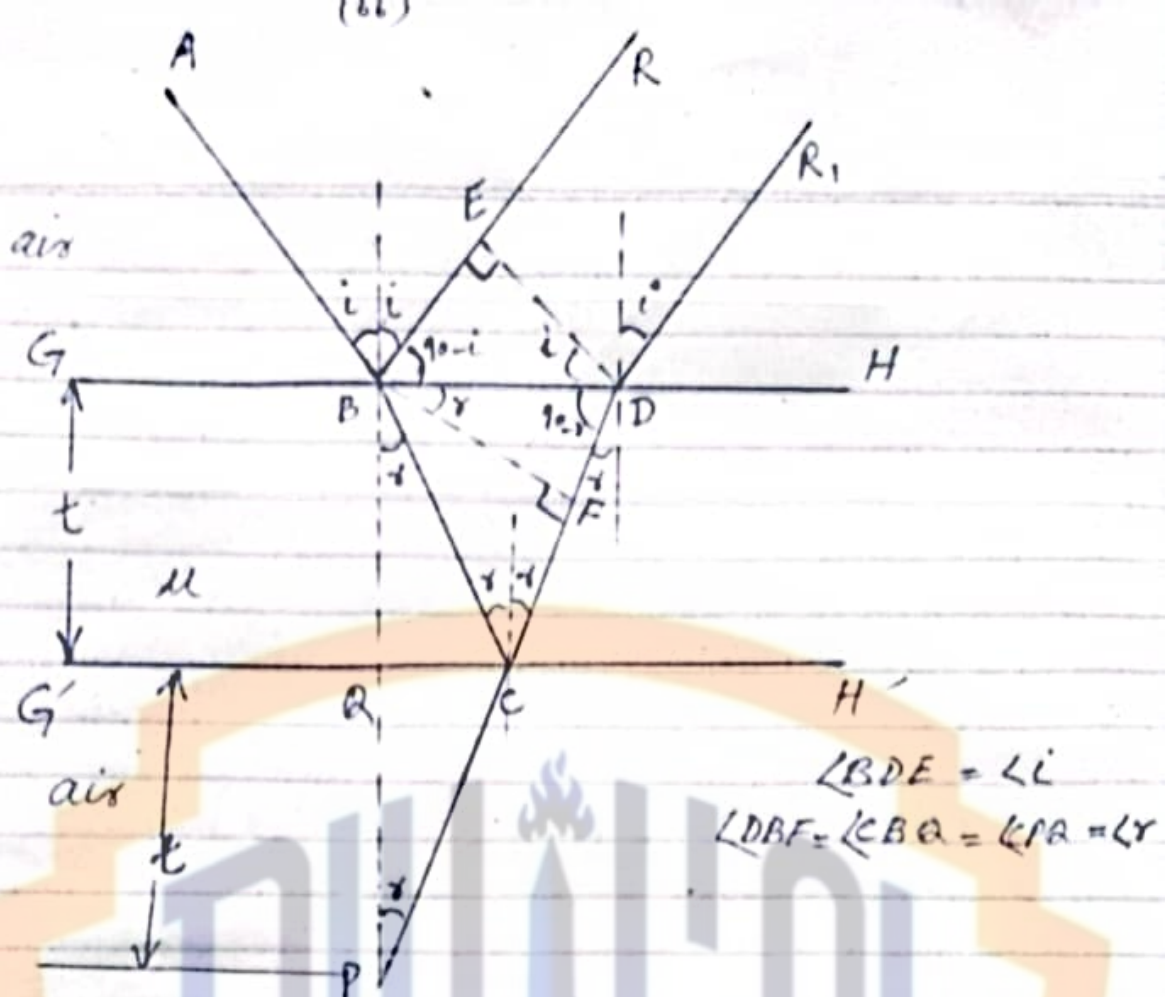
This is an example of interference due to multiple reflections from thin film.

Robert Hooke observed colours in thin film of mica. Newton produced interference rings by thin wedge shaped film b/w a convex lens and a plane glass plate. Thomas Young explained interference b/w the light rays reflected from top & bottom surfaces of a thin film.

Consider a transparent thin film of thickness " t " & refractive index " μ ".
Let GH & $G'H'$ be its parallel faces.

Suppose a light ray from a monochromatic source A is incident on GH at point B at an angle $\angle i$. A part of it is reflected back along BR . Another part of it is refracted through the film along BC at an angle $\angle r$, as shown. A part of ray BC reflected from the bottom of the film along CD , where CD is refracted along DR .

$DE \perp BR$ and $BF \perp CD$
produce CD in backward direction to meet BR at point P .



Path difference = $(BC + CD)$ in film $\rightarrow (BE)$ in air

$$\Delta = \mu(BC + CD) - 1(BE) \quad (\text{path in air} = 1 \times \text{path in medium})$$

$$\Delta = \mu(BC + CD) - (BE) \quad \text{--- (1)}$$

As $\mu = \frac{\sin i}{\sin r}$

$$\mu = \frac{BE/BD}{FD/BD} \Rightarrow \mu = \frac{BE}{FD} \Rightarrow BE = \mu(FD)$$

eq (1) becomes as:

$$\Delta = \mu(BC + CD) - \mu(FD)$$

$$\Delta = \mu(BC + CD - FD) \quad \therefore BC = PC$$

$$\Delta = \mu(PC + CD - FD)$$

$$\Delta = \mu(PC + CF + FD - FD)$$

$$\Delta = \mu(PC + CF)$$

$$\Delta = \mu \cdot PF \quad \text{--- (2)}$$

(67)

$$\text{In } \triangle BPF \quad \cos r = \frac{PF}{PB} \Rightarrow PF = PB \cos r$$

$$PF = 2BQ \cos r$$

$$PF = 2t \cos r$$

eg 6 becomes as:

$$\Delta = \mu(2t \cos r)$$

$$\Delta = 2\mu t \cos r$$

The net P.D of two reflecting waves from top & bottom of the thin film is given as:

$$P.D = 2\mu t \cos r - \lambda/2$$

Condition for Maxima:

$$P.D = n\lambda \quad \text{where } n = 0, 1, 2, 3, \dots$$

$$\Rightarrow 2\mu t \cos r - \lambda/2 = n\lambda$$

$$\Rightarrow 2\mu t \cos r = (2n+1)\lambda/2$$

Condition for Minima:-

$$P.D = (2n-1)\lambda/2$$

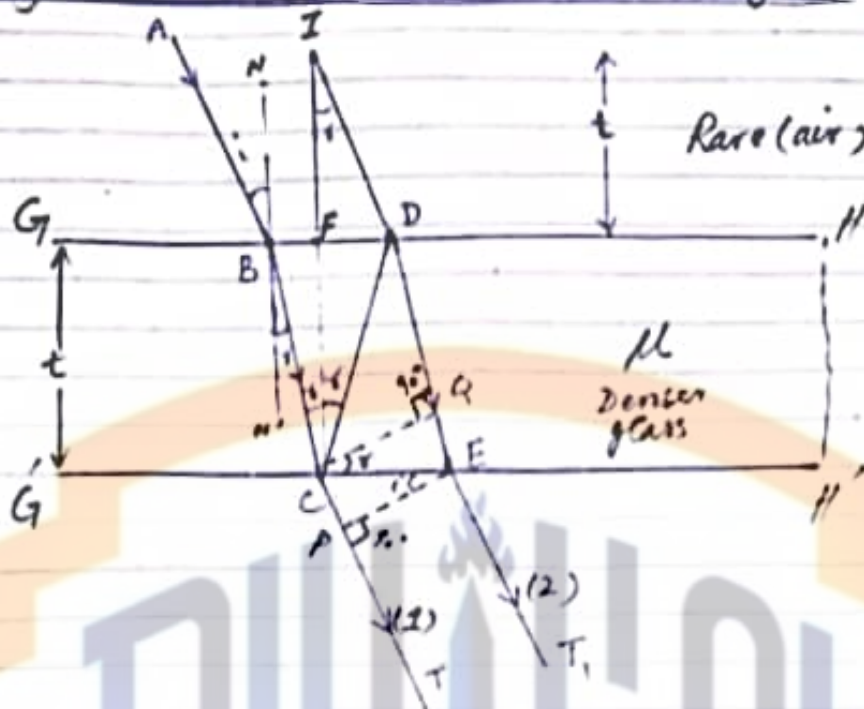
$$\Rightarrow 2\mu t \cos r - \lambda/2 = (2n-1)\lambda/2$$

$$\Rightarrow 2\mu t \cos r = \lambda/2 + 2n\lambda/2 - \lambda/2$$

$$\Rightarrow 2\mu t \cos r = n\lambda$$

(68)

Interference due to Transmitted rays:-



Incident ray AB Transmitted from the bottom of the film along CT & ray DE along ET, Draw $EP \perp CT$ & $CQ \perp DE$.

The Path difference b/w ray 1 & ray 2 = $\mu(CD + DE) - CP$

$$\Delta = \mu(CD + DE) - CP \quad \text{--- (1)}$$

In $\triangle CPE$ $\sin i = CP/CE$ & In $\triangle CQE$, $\sin r = QE/CE$

By Snell's Law: $\mu = \frac{\sin i}{\sin r}$

$$\mu = \frac{CP/CE}{QE/CE} \Rightarrow \mu = \frac{CP}{QE}$$

$$CP = \mu(QE) \quad \text{--- (2)}$$

(49)

pulling value of CP from eq in eq (1)

$$\rightarrow \Delta = \mu(CD + DE) - \mu(QE)$$

$$\Delta = \mu(CD + DE - QE)$$

$$\Delta = \mu(CD + DQ + QE - QE)$$

$$\Delta = \mu(CD + DQ)$$

$$\Delta = \mu(ID + DQ)$$

$$\Delta = \mu(IQ) \text{ --- (3)}$$

As in ΔIAC $\cos r = IQ/IC$

$$\rightarrow IQ = IC \cos r$$

$$\rightarrow IQ = (IF + FC) \cos r$$

$$\rightarrow IQ = (t + t) \cos r$$

$$\rightarrow IQ = 2t \cos r$$

using this value in eq (3)

$$\Delta = \mu(2t \cos r)$$

$$\Delta = 2\mu t \cos r$$

Net path difference $= \Delta = 2\mu t \cos r$

i) For Maxima: $\Delta = n\lambda$

$$\rightarrow \boxed{2\mu t \cos r = n\lambda} ; n = 0, 1, 2, 3, \dots$$

In this condition film will appear bright.

ii) For Minima:-

$$\Delta = (2n+1)\lambda/2$$

$$\rightarrow \boxed{2\mu t \cos r = (2n+1)\lambda/2} ; n = 0, 1, 2, 3, \dots$$

In this condition film will appear dark.

Bragg's Law:-

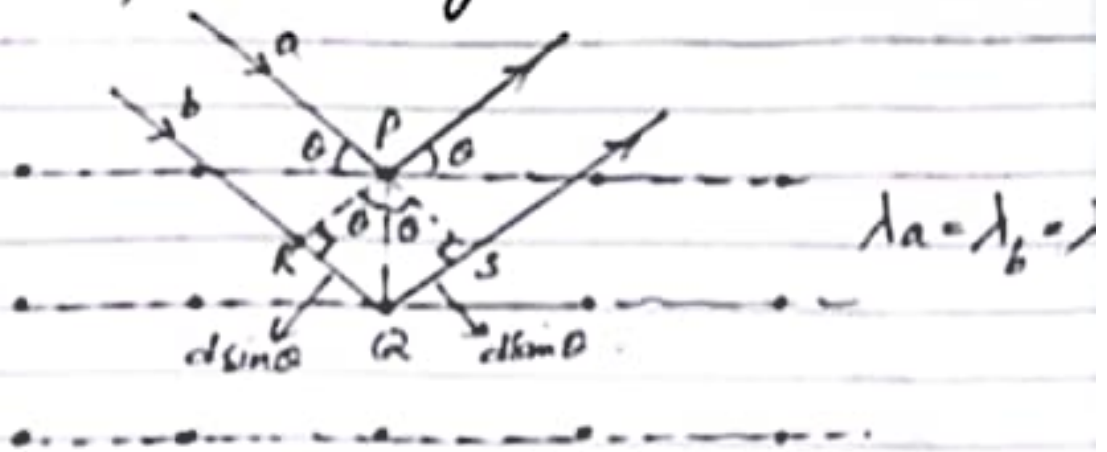
X-rays are e.m waves with very short wavelength, so their diffraction with ordinary grating is not possible. However the diffraction of X-rays by crystal is possible due to the reason that the spacing b/w the planes of atoms is comparable to the wavelength of the X-rays.

As a crystal behave like a three dimensional space grating.

In 1912, W.L. Braggs proposed a simple formulation and expression for the geometric conditions which must be satisfied. If waves are diffracted by a parallel set of planes of atoms.

Suppose two monochromatic X-rays beams a and b of wavelength " λ " are reflected from atoms P and Q located in the two

consecutive planes along the same vertical line.



Let incident beam make an angle θ with crystal plane called glancing angle. Let d be the space b/w two consecutive planes, called interplanar spacing or space grating element.

To find the path difference b/w two beams we draw lines PR and PS.

$$P.D = RQ + QS$$

$$P.D = d \sin \theta + d \sin \theta$$

$$P.D = 2d \sin \theta$$

These two // reflected beams interfere constructively & will get a bright spot on the photographic film if the path difference is equal to integral multiple of λ .

i.e.

$$2d \sin \theta = n\lambda$$

where $n = 1, 2, 3$

'n' denotes the order of the image.

Knowing d from density and atomic weight of the crystal and measuring we can find λ .

Conversely if x-rays of known wavelength are used & the with the help of Bragg's law, we can find the interplanar spacing.

The investigation of crystal structure by Bragg's Law has formed the basis of the x-ray crystallography.

($2d \sin \theta = n\lambda$ is the mathematical form of Bragg's Law)

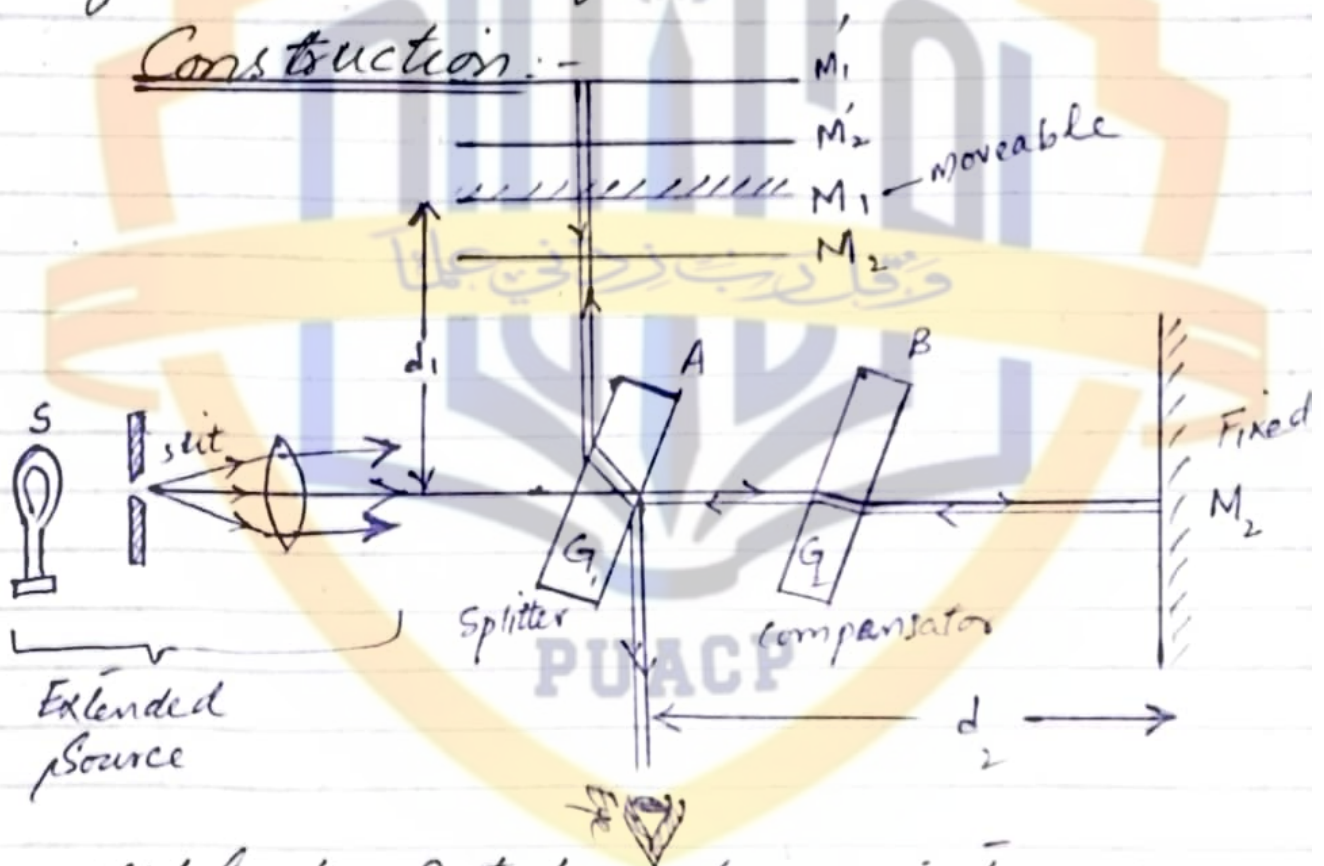
Michelson's Interferometer:-

Michelson Interferometer was discovered by Albert Abraham Michelson in the year 1881.

Principle:-

It works on the principle of interference of light by the division of amplitude in which light from an extended source is divided into two parts of equal intensity by partial reflection and refraction.

Construction:-



Michelson's Interferometer consists of a light source S . two plane mirrors M_1 & M_2 with leveling screws for adjustment of position and inclination of mirrors. Two glass plates G_1 (beam splitter) and G_2 (compensator). It also consists of a telescope to view the interference fringes.

Working :-

A light beam travels from the source S to plate G_1 , which is polished on back side. G_1 splits the beam into two parts. The reflected ray from G_1 goes to mirror M_1 (movable).

The refracted ray from G_1 goes to mirror M_2 (Fixed) by passing through compensator plate G_2 . After reflection from the mirrors M_1 & M_2 both rays form a virtual image of source S at M_1' and M_2' . M_1 and M_2 have path difference d from the G_1 and their images M_1' & M_2' have path difference $2d$.

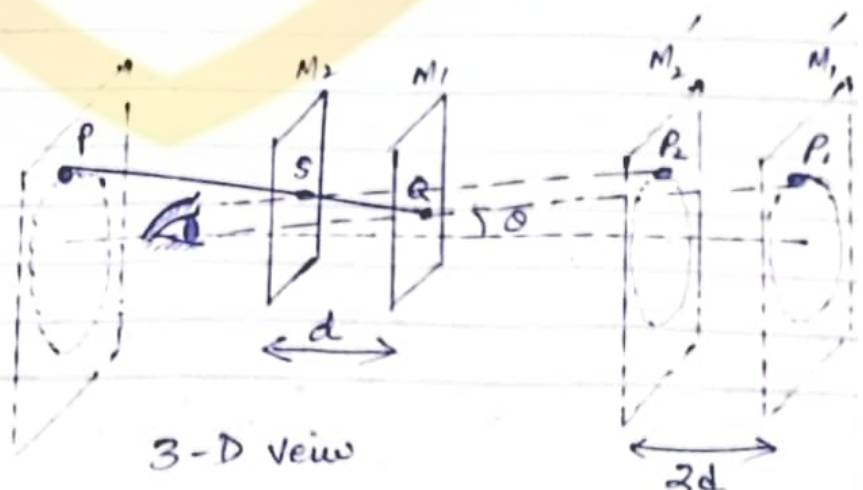
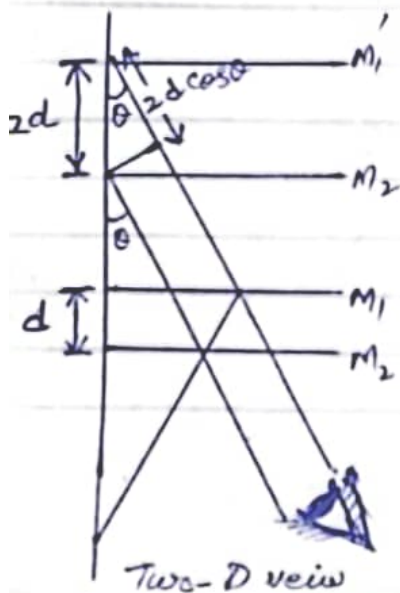
The two rays of light after reflection from the mirrors interfere constructively & destructively form bright & dark fringes which are viewed through telescope.

$$P.D = 2(d_1 - d_2) + \lambda/2$$

$$P.D = 2d + \lambda/2$$

$\lambda/2$ is additional P.D. for reflected ray according to Stoke's law.

PUACP



According to above figure The actual relation for path difference b/w reflected & refracted ray is given as:

$$PD = 2d \cos \theta + \lambda/2$$

For bright fringe:- If the two rays after reflection from mirror M_1 & M_2 interfere constructively then

$$2d \cos \theta + \lambda/2 = n\lambda$$

OR $2d \cos \theta = (n - \frac{1}{2})\lambda$ where $n = 1, 2, 3, 4, \dots$

For dark fringe:- If the two rays after reflection from mirror M_1 & mirror M_2 interfere destructively, then

$$2d \cos \theta + \lambda/2 = (n + \frac{1}{2})\lambda$$

OR $2d \cos \theta = n\lambda$ where $n = 0, 1, 2, 3, 4, \dots$

Difference of Wavelength of Spectral lines

If we use two different colours light source having different wave lengths. then suppose if first light forms n fringe at a point, then the 2nd light forms $(n+1)$ fringe at that point in such a case:

$$2d \cos \theta = n\lambda,$$

$$2d = n\lambda. \quad \text{--- (1)}$$

and $2d = (n+1)\lambda_2$ — (2)

from eq (1) & (2)

$$n\lambda_1 = (n+1)\lambda_2$$

$$n\lambda_1 = n\lambda_2 + \lambda_2$$

$$n(\lambda_1 - \lambda_2) = \lambda_2$$

$$n = \frac{\lambda_2}{\lambda_1 - \lambda_2} \text{ — (3)}$$

from eq (1)

$$2d = n\lambda_1$$

$$2d = \left(\frac{\lambda_2}{\lambda_1 - \lambda_2} \right) \lambda_1 \text{ using eq (3)}$$

$$2d = \frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2}$$

$$2d = \frac{\lambda_{av}^2}{\lambda_1 - \lambda_2}$$

$$\lambda_1 - \lambda_2 = \frac{\lambda_{av}^2}{2d}$$

let

$$\lambda_1 = \lambda_{av} + x$$

$$\lambda_2 = \lambda_{av} - x$$

$$\lambda_1 \lambda_2 = \lambda_{av}^2 - x^2$$

$$\lambda_1 \lambda_2 \approx \lambda_{av}^2 \text{ if } x \ll \lambda_{av}$$

So the difference of wavelength of two light can be calculated by using above